

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A150NRL31HN001Ammonia accumulator7%.pvdb

Date of Analysis : Sep 15, 2026 4:02pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2025.000 mm ID
 Vessel Design Length, Tangent to Tangent 820.00 cm
 Specified Datum Line Distance 0.00 cm
 Internal Design Temperature 85 °C
 Internal Design Pressure 25.492 kgf/cm²
 External Design Temperature 85 °C
 Maximum Allowable Working Pressure 28 kgf/cm²
 Shop Test Pressure 36.411 kgf/cm²
 Field Test Pressure 36.411 kgf/cm²
 Required Minimum Design Metal Temperature -40.0 °C
 Warmest Computed Minimum Design Metal Temperature -29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code ASCE/SEI 7-22
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+N+IT
Head	SA-516 70	...	...	K02700	+N+IT
Nozzle	SA-350 LF1	1	...	K03009	+IT
Nozzle Flg	SA-350 LF1	1	...	K03009	+IT
Nozzle Flg		...	...		
Hrz Bolting		...	...		
Saddles	SA-285 C	...	...	K02801	+N

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Head-1	25.591	0.00	34.000	1.5000	No	No
Shell-1	25.591	0.00	28.000	1.5000	N/A	No
Shell-2	25.591	0.00	28.000	1.5000	N/A	No
Shell-3	25.591	0.00	28.000	1.5000	N/A	No
Head-2	25.591	0.00	28.000	1.5000	No	No

Liquid Level: 170.00 cm Dens.: 579.734 kg/m³ Sp. Gr.: 0.580

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	5.0	5.0	32.0	26.0	19.9	...	1.00	1.00
Cylinder	305.0	300.0	22.0	22.0	20.1	...	1.00	1.00
Cylinder	605.0	300.0	22.0	22.0	20.1	...	1.00	1.00
Cylinder	815.0	210.0	22.0	22.0	20.1	...	1.00	1.00
Ellipse	820.0	5.0	26.0	22.0	19.9	...	1.00	1.00

Local Stress Analysis Results:

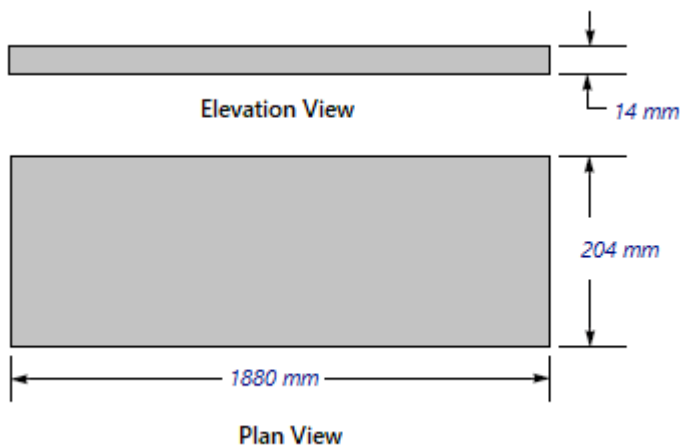
Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
A1_6"	WRC-297	0.994	Shell	Passed
A2_6"	WRC-297	0.994	Shell	Passed
A3_2"	WRC-297	0.998	Shell	Passed
S1_3"	WRC-297	0.994	Shell	Passed
V1_3"	WRC-297	0.994	Shell	Passed
T1_2"	WRC-297	0.998	Shell	Passed
B2_6"	WRC-297	0.998	Shell	Passed
B1_3"	WRC-297	0.994	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	178.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	1550.000	mm
Wear Pad Width	230.000	mm
Wear Pad Thickness	10.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	1000.000	mm
Baseplate Length	1880.000	mm
Baseplate Thickness	14.000	mm
Baseplate Width	204.000	mm
Number of Ribs (including outside ribs)	3	
Rib Thickness	14.000	mm
Web Thickness	14.000	mm
Height of Center Web	480.000	mm
Number of Bolts in Baseplate	4	

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	25122.51	kgf
Maximum Transverse Saddle Shear Load	2739.64	kgf
Maximum Longitudinal Saddle Shear Load	4729.57	kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	26744.94	kgf
Maximum Transverse Saddle Shear Load	904.08	kgf
Maximum Longitudinal Saddle Shear Load	585.28	kgf

Weights:



Fabricated - Bare w/o removable internals	14580.6	kg
Shop Test - Fabricated + water (full)	43150.8	kg
Shipping - Fab. + rem. intls.+ shipping app.	17771.7	kg
Erected - Fab. + rem. intls.+ insul. (etc)	17971.7	kg
Empty - Fab. + intls. + details + wghts.	17971.7	kg
Operating - Empty + operating liquid (no ca)	32873.7	kg
Field Test - Empty weight + test medium (full)	46341.9	kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
M1_24"	28	...	...	...		OK	Passed
U1_8"	31	...	...	OK	8.66	OK	Passed
U1_8"	34	...	...	OK	8.66	OK	Passed
A1_6"	28	...	...	OK	7.72	OK	Passed
A2_6"	28	...	...	OK	7.72	OK	Passed
P1_2"	28	...	...	OK	4.92	OK	No Calc[*]
P2_2"	28	...	...	OK	4.92	OK	No Calc[*]
A3_2"	28	...	...	OK	4.92	OK	No Calc[*]
L3_2"	28	...	...	OK	4.92	OK	No Calc[*]
L4_2"	28	...	...	OK	4.92	OK	No Calc[*]
S1_3"	28	...	...	OK	6.30	OK	Passed
V1_3"	28	...	...	OK	6.30	OK	Passed
T1_2"	28	...	...	OK	4.92	OK	No Calc[*]
L1_2"	28	...	...	OK	4.92	OK	No Calc[*]
L2_2"	28	...	...	OK	4.92	OK	No Calc[*]
B2_6"	28	...	...	OK	7.72	OK	Passed
B1_3"	28	...	...	OK	6.30	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 28 Nozzle : B13"
 Minimum MAWP Shells/Flanges : 28 Element : Shell-3
 Minimum MAPnc Shells/Flanges : 30 Element : Shell-3

Computed Vessel M.A.W.P. : 28 kgf/cm²

[*] - This was a small opening and the areas were not computed.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	M1_24"	0.000	0.000	1129.200
10	U1_8"	0.000	270.000	399.700
20	A1_6"	500.000	90.000	304.145
20	A2_6"	1400.000	90.000	304.145
20	P1_2"	2050.000	90.000	134.034
20	P2_2"	2550.000	90.000	134.034
20	A3_2"	2050.000	270.000	134.034
20	L3_2"	1550.000	28.903	134.034
20	L4_2"	1550.000	-28.903	134.034
30	S1_3"	4550.000	90.000	144.862
30	V1_3"	5050.000	90.000	144.862
30	T1_2"	4050.000	270.000	134.034
30	L1_2"	4550.000	28.903	134.034
30	L2_2"	4550.000	-28.903	134.034
40	B2_6"	7550.001	270.000	304.145
40	B1_3"	7550.001	90.000	158.431

The nozzle spacing is computed by the following:

$$= \sqrt{(l_l^2 + l_c^2)} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!



Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
P1_2"	2.000 in	160	LWN	2.375	8.738	...	...	423.07	300
P2_2"	2.000 in	160	LWN	2.375	8.738	...	...	423.07	300
A3_2"	2.000 in	160	LWN	2.375	8.738	...	...	223.07	600
L3_2"	2.000 in	160	LWN	2.375	8.738	...	...	253.31	300
L4_2"	2.000 in	160	LWN	2.375	8.738	...	...	253.31	300
T1_2"	2.000 in	160	LWN	2.375	8.738	...	...	423.07	600
L1_2"	2.000 in	160	LWN	2.375	8.738	...	...	232.73	300
L2_2"	2.000 in	160	LWN	2.375	8.738	...	...	232.73	300
S1_3"	3.000 in	160	WNF	3.500	11.125	...	...	423.70	600
V1_3"	3.000 in	160	WNF	3.500	11.125	...	...	423.70	600
B1_3"	3.000 in	160	WNF	3.500	11.125	...	...	423.70	600
A1_6"	6.000 in	80	WNF	6.625	10.973	...	...	428.96	600
A2_6"	6.000 in	80	WNF	6.625	10.973	...	...	428.96	600
B2_6"	6.000 in	80	WNF	6.625	10.973	...	...	228.96	300
U1_8"	8.000 in	80	WNF	8.625	12.700	...	...	311.29	300
M1_24"	24.000 in Actual		WNF	24.000	24.000	...	...	351.37	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:
 Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
P1_2"	SA-350 LF1	22.000	10.000	...	...	...
P2_2"	SA-350 LF1	22.000	10.000	...	...	...
A3_2"	SA-350 LF1	22.000	10.000	...	...	...
L3_2"	SA-350 LF1	22.000	10.000	...	...	...
L4_2"	SA-350 LF1	22.000	10.000	...	...	...
T1_2"	SA-350 LF1	22.000	10.000	...	...	...
L1_2"	SA-350 LF1	22.000	10.000	...	...	...
L2_2"	SA-350 LF1	22.000	10.000	...	...	...
S1_3"	SA-350 LF1	22.000	10.000	...	...	...
V1_3"	SA-350 LF1	22.000	10.000	...	...	...
B1_3"	SA-350 LF1	22.000	10.000	...	...	...
A1_6"	SA-350 LF1	22.000	10.000	...	...	...
A2_6"	SA-350 LF1	22.000	10.000	...	...	...
B2_6"	SA-350 LF1	22.000	10.000	...	...	...
U1_8"	SA-350 LF1	20.000	10.000	...	...	...
M1_24"	SA-350 LF1	20.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
P1_2"	205.000	90.0	400.00	0.00	Shell-1
P2_2"	255.000	90.0	400.00	0.00	Shell-1
A3_2"	205.000	270.0	200.00	0.00	Shell-1
L3_2"	155.000	28.9	200.00	0.00	Shell-1
L4_2"	155.000	-28.9	200.00	0.00	Shell-1



T1_2"	405.000	270.0	400.00	0.00	Shell-2
L1_2"	455.000	28.9	179.43	0.00	Shell-2
L2_2"	455.000	-28.9	179.43	0.00	Shell-2
S1_3"	455.000	90.0	400.00	0.00	Shell-2
V1_3"	505.000	90.0	400.00	0.00	Shell-2
B1_3"	755.000	90.0	400.00	0.00	Shell-3
A1_6"	50.000	90.0	400.00	0.00	Shell-1
A2_6"	140.000	90.0	400.00	0.00	Shell-1
B2_6"	755.000	270.0	200.00	0.00	Shell-3
U1_8"	...	270.0	200.00	0.00	Head-1
M1_24"	...	0.0	300.00	0.00	Head-1

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 32.0mm THK X 2025.0mm ID X 5.0cm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 26.0mm THK X 2025.0mm ID X 5.0cm	SA-516 70
2	SADDLE: 178.0mm X 120 DEG	SA-285 C
1	PLATFORM: 1100mm X 2100mm WIDE	
1	LADDER: 2500.0mm	...
1	CLASS 300 GR 1.4, 24.0" BLIND FLANGE(S)	SA-350 LF1
3	CYLINDRICAL SEGMENT 22.0mm THK X 3048.0mm X 300.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 265.7mm X 300.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 3313.7mm X 300.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 0.0mm X 300.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 3048.0mm X 210.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 3313.7mm X 210.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 0.0mm X 210.0cm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Head-1	[10]	!	-40	-50		0.827	26.000	1.00	No
Head-1	[7]	!	-40	-58		0.672	32.000	1.00	No
Shell-1	[8]	!	-40	-40		1.000	22.000	1.00	No
Shell-2	[8]	!	-40	-40		1.000	22.000	1.00	No
Shell-3	[8]	!	-40	-40		1.000	22.000	1.00	No
Head-2	[10]	!	-40	-41		0.988	22.000	1.00	No
Head-2	[7]	!	-40	-49		0.837	26.000	1.00	No
M124"	[1]	!	-29	-29	-29	0.743	26.000	1.00	No
Nozzle Flg	[5]	!	-29	-49					
U18"	[1]	!	-29	-29	-29	0.826	26.000	1.00	No
Nozzle Flg	[5]	!	-29	-49					
A16"	[1]	!	-29	-29	-29	0.996	22.000	1.00	No
Nozzle Flg	[5]	!	-29	-104					
A26"	[1]	!	-29	-29	-29	0.996	22.000	1.00	No
Nozzle Flg	[5]	!	-29	-104					
P12"	[1]	!	-29	-29	-29	0.996	22.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
P22"	[1]	!	-29	-29	-29	0.996	22.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
A32"	[1]	!	-29	-29	-29	1.000	22.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
L32"	[1]	!	-29	-29	-29	0.997	22.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
L42"	[1]	!	-29	-29	-29	0.999	22.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
S13"	[1]	!	-29	-29	-29	0.996	22.000	1.00	No
Nozzle Flg	[5]	!	-29	-104					
V13"	[1]	!	-29	-29	-29	0.996	22.000	1.00	No
Nozzle Flg	[5]	!	-29	-104					
T12"	[1]	!	-29	-29	-29	1.000	22.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
L12"	[1]	!	-29	-29	-29	0.997	22.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
L22"	[1]	!	-29	-29	-29	0.999	22.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
B26"	[1]	!	-29	-29	-29	1.000	22.000	1.00	No
Nozzle Flg	[5]	!	-29	-57					
B13"	[1]	!	-29	-29	-29	0.996	22.000	1.00	No
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: -29 -29

Required Minimum Design Metal Temperature -40.0 °C
 Warmest Computed Minimum Design Metal Temperature -29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material



- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfc^{m2}):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in. ⁴	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in. ³	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	25.492	kgf/cm ²
Design Internal Temperature	85.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-40.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW	
Load Case 17	NP+VO+OW	
Load Case 18	FS+BS+IP+OW	
Load Case 19	FS+BS+EP+OW	
Load Case 20	BL+IP+OW	
Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	Moderately Smooth	
Base Elevation	460	cm
Percent Wind for Hydrotest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh 0	cm
Distance Upwind of Crest	Lh 0	cm
Distance from Crest to the Vessel	x 0	cm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	cm
Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio	z/h 0.500	
Amplification Factor	Ap 1.000	
Force Factor	0.000	
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Component resonance duct. fact.	[CAR] 1.000	
Component strength factor	[Rpo] 1.500	
Overstrength factor	[Omega0] 2.000	
Response Acceleration Factor	[SMS] 0.326	
Response Acceleration Factor	[SM1] 0.139	
Building Importance Factor	[Ie] 1.000	
Site Latitude	0.000	deg
Site Longitude	0.000	deg
M.A.W.P + Static Head to Nozzle	Y	
Consider MAP New and Cold in Noz. Design	N	
Consider External Loads for Nozzle Des.	Y	
Use ASME VIII-1 Appendix 1-9	N	
Perform Blast Load Analysis	No	
Material Database Year	Current w/Addenda or Code Year	

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:



Element#	1/5
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Head-1
Distance "FROM" to "TO"	5 cm
Inside Diameter	2025 mm
Element Thickness	26 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	32 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	25.492 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Allowable Stress, Ambient	1407.2 kgf/cm ²
Allowable Stress, Operating	1407.2 kgf/cm ²
Allowable Stress, Hydrotest	2386.1 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	2467.6 kgf/cm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	170 cm
Liquid Density	579.73 kg/m ³

Element From Node	10
Detail Type	Nozzle
Detail ID	M124"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	855.21 kgf
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-350 LF1 +IT

Element From Node	10
Detail Type	Nozzle
Detail ID	U18"
Dist. from "FROM" Node / Offset dist	800 mm
Nozzle Diameter	8 in.
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	79.671 kgf
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-350 LF1 +IT

Element#	2/5
Element From Node	20
Element To Node	30



Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	300 cm
Inside Diameter	2025 mm
Element Thickness	22 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	25.492 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	20
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	95 cm
Width of Saddle	178 mm
Height of Saddle at Bottom	1550 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	230 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.30000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 cm
Element From Node	20
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	0 cm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	188192 cm ²
Platform Weight	2982.3 kgf
Platform Railing Weight	0 kgf/cm
Platform Grating Weight	0.017 kgf/cm ²
Platform Width	2100 mm
Platform Height	1100 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	2.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 cm
Ladder End Elevation	250 cm
Unit Weight of Ladder	0.4997 kgf/cm
Platform Length (top head platform)	8000 mm
Element From Node	20
Detail Type	Liquid
Detail ID	Liquid 20
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	170 cm
Liquid Density	579.73 kg/m ³
Element From Node	20
Detail Type	Nozzle
Detail ID	A16"
Dist. from "FROM" Node / Offset dist	45 cm
Nozzle Diameter	6 in.
Nozzle Schedule	80
Nozzle Class	600



Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	55.276	kgf
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	A26"	
Dist. from "FROM" Node / Offset dist	135	cm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	55.276	kgf
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	P12"	
Dist. from "FROM" Node / Offset dist	200	cm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	11.91	kgf
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	P22"	
Dist. from "FROM" Node / Offset dist	250	cm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	11.476	kgf
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	A32"	
Dist. from "FROM" Node / Offset dist	200	cm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.5962	kgf
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	L32"	
Dist. from "FROM" Node / Offset dist	150	cm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	28.9028	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.5962	kgf
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	



Element From Node	20
Detail Type	Nozzle
Detail ID	L42"
Dist. from "FROM" Node / Offset dist	150 cm
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	-28.9028
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	9.5962 kgf
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-350 LF1 +IT

Element From Node	20
Detail Type	Weight
Detail ID	Lifting lug
Dist. from "FROM" Node / Offset dist	50 cm
Miscellaneous Weight	420 kgf
Offset from Element Centerline	0 mm

Element From Node	20
Detail Type	Weight
Detail ID	HalfopenpipeA
Dist. from "FROM" Node / Offset dist	100 cm
Miscellaneous Weight	200 kgf
Offset from Element Centerline	0 mm

Element#	3/5
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Shell-2
Distance "FROM" to "TO"	300 cm
Inside Diameter	2025 mm
Element Thickness	22 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	25.492 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	170 cm
Liquid Density	579.73 kg/m ³

Element From Node	30
Detail Type	Nozzle
Detail ID	S13"
Dist. from "FROM" Node / Offset dist	150 cm
Nozzle Diameter	3 in.
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	19.574 kgf
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-350 LF1 +IT



Element From Node 30
 Detail Type Nozzle
 Detail ID V13"
 Dist. from "FROM" Node / Offset dist 200 cm
 Nozzle Diameter 3 in.
 Nozzle Schedule 160
 Nozzle Class 600
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 20.846 kgf
 Grade of Attached Flange GR 1.4
 Nozzle Matl SA-350 LF1 +IT

Element From Node 30
 Detail Type Nozzle
 Detail ID T12"
 Dist. from "FROM" Node / Offset dist 100 cm
 Nozzle Diameter 2 in.
 Nozzle Schedule 160
 Nozzle Class 600
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 12.223 kgf
 Grade of Attached Flange GR 1.4
 Nozzle Matl SA-350 LF1 +IT

Element From Node 30
 Detail Type Nozzle
 Detail ID L12"
 Dist. from "FROM" Node / Offset dist 150 cm
 Nozzle Diameter 2 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle 28.9028
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 9.5962 kgf
 Grade of Attached Flange GR 1.4
 Nozzle Matl SA-350 LF1 +IT

Element From Node 30
 Detail Type Nozzle
 Detail ID L22"
 Dist. from "FROM" Node / Offset dist 150 cm
 Nozzle Diameter 2 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle -28.9028
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 9.5962 kgf
 Grade of Attached Flange GR 1.4
 Nozzle Matl SA-350 LF1 +IT

Element From Node 30
 Detail Type Weight
 Detail ID Clipwt
 Dist. from "FROM" Node / Offset dist 0 cm
 Miscellaneous Weight 81.999 kgf
 Offset from Element Centerline 0 mm

Element# 4/5
 Element From Node 40
 Element To Node 50
 Element Type Cylinder
 Description Shell-3
 Distance "FROM" to "TO" 210 cm
 Inside Diameter 2025 mm
 Element Thickness 22 mm



Internal Corrosion Allowance	1.5	mm
Nominal Thickness	22	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+N+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	New Saddle	
Dist. from "FROM" Node / Offset dist	115	cm
Width of Saddle	178	mm
Height of Saddle at Bottom	1550	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	230	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid 40	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	170	cm
Liquid Density	579.73	kg/m ³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	B26"	
Dist. from "FROM" Node / Offset dist	150	cm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	58.076	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF1	+IT

Element From Node	40	
Detail Type	Nozzle	
Detail ID	B13"	
Dist. from "FROM" Node / Offset dist	150	cm
Nozzle Diameter	3	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	11.91	kgf
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1	+IT

Element#	5/5
Element From Node	50
Element To Node	60
Element Type	Elliptical



Description	Head-2	
Distance "FROM" to "TO"	5	cm
Inside Diameter	2025	mm
Element Thickness	22	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	26	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+N+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid 50	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	170	cm
Liquid Density	579.73	kg/m ³

XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
Head-1		5	...	5	...
Shell-1		305	...	300	...
Shell-2		605	...	300	...
Shell-3		815	...	210	...
Head-2		820	...	5	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
Head-1		25.591	32	1.5	2025	1407.2	1
Shell-1		25.591	22	1.5	2025	1407.1	2
Shell-2		25.591	22	1.5	2025	1407.1	3
Shell-3		25.591	22	1.5	2025	1407.1	4
Head-2		25.591	26	1.5	2025	1407.1	5

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
Head-1		25.492	33.8855	36.0419	26	19.9379	1
Shell-1		25.492	28.0084	30.1813	22	20.1444	2
Shell-2		25.492	28.0084	30.1813	22	20.1444	3
Shell-3		25.492	28.0084	30.1813	22	20.1444	4
Head-2		25.492	28.3478	30.5085	22	19.9382	5

Minimum 28 30

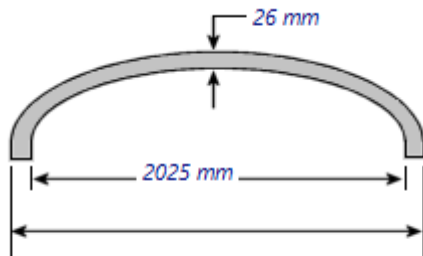
MAWP: 28 kgf/cm², limited by: Shell-3.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Head-1

Elliptical Head Design Information:							
Design Pressure	P	25.591	kgf/cm²	Int. Design Temperature		85	°C
Material		SA-516	70	Ext. Design Temperature		85	°C
Length		5.000	cm	External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.	c	1.5000	mm	Allowable Stress (ope)	S	1407.2	kgf/cm²
Ext. Corr. All.	ce	0.0000	mm	Allowable Stress (amb)	Sa	1407.2	kgf/cm²
Inside Diameter	Di	2025.000	mm	Actual Stress		1059.6	kgf/cm²
Spec. Min. Thk t _e , or s _a		26.0000	mm	Spec. Nominal Thk.		32.000	mm
Surface Area		50585.7	cm²	SG of Contents		0.580	
Element Volume		1247994.5	cm³	Weight of Contents		673.6	kgf
Empty Weight		1336.0	kgf	Operating Weight		2009.6	kgf
				Comp. All. Ope. Stress		1365.6	kgf/cm²
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 25.312 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 12.656 mm

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (25.591 \cdot 2028.0 \cdot 0.998) / (2 \cdot 1407.16 \cdot 1.0 - 0.2 \cdot 25.591)$$

$$= 18.4379 + 1.5000 = 19.9379 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.099 kgf/cm²

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.16 \cdot 1.0 \cdot 24.5) / (0.998 \cdot 2028.0 + 0.2 \cdot 24.5)$$

$$= 34 - 0.1 = 34 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.16 \cdot 1.0 \cdot 26.0) / (1.0 \cdot 2025.0 + 0.2 \cdot 26.0)$$

$$= 36 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (25.591 \cdot (0.998 \cdot 2028.0 + 0.2 \cdot 24.5)) / (2 \cdot 1.0 \cdot 24.5)$$



= 1059.615 kgf/cm²

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (25.591 \cdot 1014.0) / (1407.16 \cdot 1.0 - 0.6 \cdot 25.591) + 1.5 \\ &= 20.144 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.099 kgf/cm²

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (1407.16 \cdot 1.0 \cdot 30.5) / (1014.0 + 0.6 \cdot 30.5) \\ &= 42 - 0.1 = 41 \text{ kgf/cm}^2 \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ &= (2 + (2028.0 / (2 \cdot 507.75))^2) / 6 \\ &= 0.998032 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o)$ 6.662 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

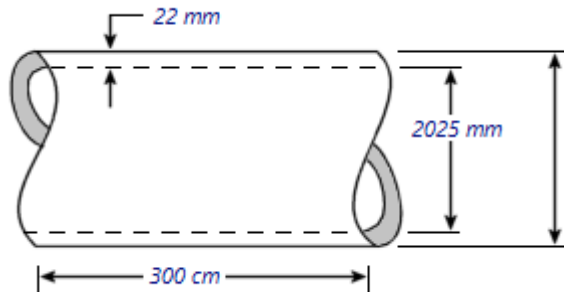
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 25.312 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 12.656 mm

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	25.591 kgf/cm ²	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		300.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm ²
Inside Diameter	Di	2025.000 mm	Actual Stress	1281.2 kgf/cm ²	
Spec. Min. Thk t _e , or s _a		22.0000 mm	Spec. Nominal Thk.	22.000 mm	
Surface Area		194998.7 cm ²	SG of Contents	0.580	
Element Volume		9661893.0 cm ³	Weight of Contents	5020.3 kgf	
Empty Weight		3289.4 kgf	Operating Weight	8309.7 kgf	
			Comp. All. Ope. Stress	1325.2 kgf/cm ²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.591 \cdot 1014.0) / (1407.13 \cdot 1.0 - 0.6 \cdot 25.591)$$

$$= 18.6444 + 1.5000 = 20.1444 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.099 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 20.5) / (1014.0 + 0.6 \cdot 20.5)$$

$$= 28 - 0.1 = 28 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 22.0) / (1012.5 + 0.6 \cdot 22.0)$$

$$= 30 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (25.591 \cdot (1014.0 + 0.6 \cdot 20.5)) / (1.0 \cdot 20.5)$$

$$= 1281.156 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.075 %

Minimum Design Metal Temperature Results:



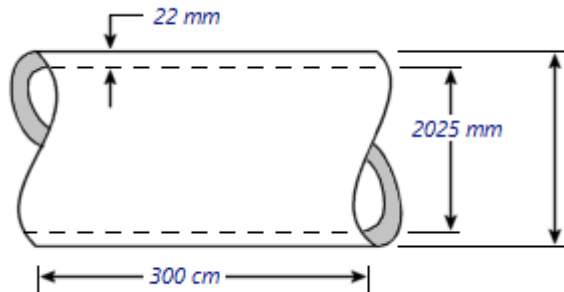
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	25.591 kgf/cm²	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		300.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	2025.000 mm	Actual Stress	1281.2 kgf/cm²	
Spec. Min. Thk t _e , or s _a		22.0000 mm	Spec. Nominal Thk.	22.000 mm	
Surface Area		194998.7 cm²	SG of Contents	0.580	
Element Volume		9661893.0 cm³	Weight of Contents	5020.3 kgf	
Empty Weight		3289.4 kgf	Operating Weight	8309.7 kgf	
			Comp. All. Ope. Stress	1325.2 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.591 \cdot 1014.0) / (1407.13 \cdot 1.0 - 0.6 \cdot 25.591)$$

$$= 18.6444 + 1.5000 = 20.1444 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.099 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 20.5) / (1014.0 + 0.6 \cdot 20.5)$$

$$= 28 - 0.1 = 28 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 22.0) / (1012.5 + 0.6 \cdot 22.0)$$

$$= 30 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (25.591 \cdot (1014.0 + 0.6 \cdot 20.5)) / (1.0 \cdot 20.5)$$

$$= 1281.156 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.075 %

Minimum Design Metal Temperature Results:



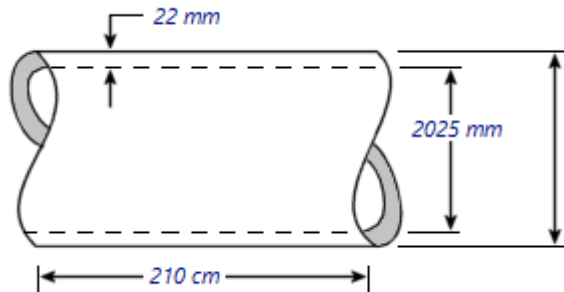
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	25.591 kgf/cm²	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		210.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	2025.000 mm	Actual Stress	1281.2 kgf/cm²	
Spec. Min. Thk t _e , or s _a		22.0000 mm	Spec. Nominal Thk.	22.000 mm	
Surface Area		136499.1 cm²	SG of Contents	0.580	
Element Volume		6763325.0 cm³	Weight of Contents	3514.2 kgf	
Empty Weight		2302.6 kgf	Operating Weight	5816.8 kgf	
			Comp. All. Ope. Stress	1325.2 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.591 \cdot 1014.0) / (1407.13 \cdot 1.0 - 0.6 \cdot 25.591)$$

$$= 18.6444 + 1.5000 = 20.1444 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.099 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 20.5) / (1014.0 + 0.6 \cdot 20.5)$$

$$= 28 - 0.1 = 28 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 22.0) / (1012.5 + 0.6 \cdot 22.0)$$

$$= 30 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (25.591 \cdot (1014.0 + 0.6 \cdot 20.5)) / (1.0 \cdot 20.5)$$

$$= 1281.156 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.075 %

Minimum Design Metal Temperature Results:

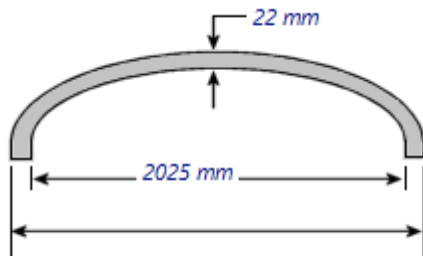
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Head-2

Elliptical Head Design Information:							
Design Pressure	P	25.591	kgf/cm²	Int. Design Temperature		85	°C
Material		SA-516	70	Ext. Design Temperature		85	°C
Length		5.000	cm	External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.	c	1.5000	mm	Allowable Stress (ope)	S	1407.1	kgf/cm²
Ext. Corr. All.	ce	0.0000	mm	Allowable Stress (amb)	Sa	1407.1	kgf/cm²
Inside Diameter	Di	2025.000	mm	Actual Stress		1265.9	kgf/cm²
Spec. Min. Thk t,e, or sa		22.0000	mm	Spec. Nominal Thk.		26.000	mm
Surface Area		50024.9	cm²	SG of Contents		0.580	
Element Volume		1247994.5	cm³	Weight of Contents		673.6	kgf
Empty Weight		1079.4	kgf	Operating Weight		1753.0	kgf
				Comp. All. Ope. Stress		1325.2	kgf/cm²
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 25.312 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 12.656 mm

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (25.591 \cdot 2028.0 \cdot 0.998) / (2 \cdot 1407.13 \cdot 1.0 - 0.2 \cdot 25.591)$$

$$= 18.4382 + 1.5000 = 19.9382 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.099 kgf/cm²

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.13 \cdot 1.0 \cdot 20.5) / (0.998 \cdot 2028.0 + 0.2 \cdot 20.5)$$

$$= 28 - 0.1 = 28 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.13 \cdot 1.0 \cdot 22.0) / (1.0 \cdot 2025.0 + 0.2 \cdot 22.0)$$

$$= 31 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (25.591 \cdot (0.998 \cdot 2028.0 + 0.2 \cdot 20.5)) / (2 \cdot 1.0 \cdot 20.5)$$

= 1265.870 kgf/cm²

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (25.591 \cdot 1014.0) / (1407.13 \cdot 1.0 - 0.6 \cdot 25.591) + 1.5 \\ &= 20.144 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.099 kgf/cm²

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (1407.13 \cdot 1.0 \cdot 24.5) / (1014.0 + 0.6 \cdot 24.5) \\ &= 34 - 0.1 = 33 \text{ kgf/cm}^2 \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6 \\ &= (2 + (2028.0 / (2 \cdot 507.75)))^2 / 6 \\ &= 0.998032 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.458 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 25.312 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 12.656 mm

Material treatment options: +N+IT

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	36.411 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	33.140 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	39.033 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	30.809 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	36.454 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	36.454 kgf/cm ²
App 27-4	= M.A.W.P.	28.008 kgf/cm ²

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 28.008 • 1.0
 = 36.411 kgf/cm²

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Head-1	1429.5	2386.1	0.599	36.61
Shell-1	1707.0	2386.0	0.715	36.61
Shell-2	1707.0	2386.0	0.715	36.61
Shell-3	1707.0	2386.0	0.715	36.61
Head-2	1688.7	2386.0	0.708	36.61

Theoretical Maximum Allowable test pressure is : 50.975 kgf/cm²

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1_24"	Nozzle	1203.20	1203.20	1.000
U1_8"	Nozzle	1203.20	1203.20	1.000
A1_6"	Nozzle	1203.20	1203.20	1.000
A2_6"	Nozzle	1203.20	1203.20	1.000
P1_2"	Nozzle	1203.20	1203.20	1.000
P2_2"	Nozzle	1203.20	1203.20	1.000
A3_2"	Nozzle	1203.20	1203.20	1.000
L3_2"	Nozzle	1203.20	1203.20	1.000
L4_2"	Nozzle	1203.20	1203.20	1.000
S1_3"	Nozzle	1203.20	1203.20	1.000
V1_3"	Nozzle	1203.20	1203.20	1.000
T1_2"	Nozzle	1203.20	1203.20	1.000
L1_2"	Nozzle	1203.20	1203.20	1.000
L2_2"	Nozzle	1203.20	1203.20	1.000
B2_6"	Nozzle	1203.20	1203.20	1.000
B1_3"	Nozzle	1203.20	1203.20	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Head-1	1407.16	1407.16	1.000
Shell-1	1407.13	1407.13	1.000
Shell-2	1407.13	1407.13	1.000
Shell-3	1407.13	1407.13	1.000
Head-2	1407.13	1407.13	1.000

Minimum 1.000



Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
M1_24"	450.34	1881.28	0.239
U1_8"	346.26	1881.28	0.184
A1_6"	306.21	1881.28	0.163
A2_6"	306.21	1881.28	0.163
P1_2"	129.80	1881.28	0.069
P2_2"	129.80	1881.28	0.069
A3_2"	129.80	1881.28	0.069
L3_2"	129.80	1881.28	0.069
L4_2"	129.80	1881.28	0.069
S1_3"	152.54	1881.28	0.081
V1_3"	152.54	1881.28	0.081
T1_2"	129.80	1881.28	0.069
L1_2"	129.80	1881.28	0.069
L2_2"	129.80	1881.28	0.069
B2_6"	306.21	1881.28	0.163
B1_3"	152.54	1881.28	0.081

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	36.411 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	33.140 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	39.033 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	30.809 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	36.454 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	36.454 kgf/cm ²
App 27-4	= M.A.W.P.	28.008 kgf/cm ²

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 28.008 • 1.0
 = 36.411 kgf/cm²

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Head-1	1516.0	2386.1	0.635	36.61
Shell-1	1833.0	2386.0	0.768	36.61
Shell-2	1833.0	2386.0	0.768	36.61
Shell-3	1833.0	2386.0	0.768	36.61
Head-2	1811.1	2386.0	0.759	36.61

Theoretical Maximum Allowable test pressure is : 47.457 kgf/cm²

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1_24"	Nozzle	1203.20	1203.20	1.000
U1_8"	Nozzle	1203.20	1203.20	1.000
A1_6"	Nozzle	1203.20	1203.20	1.000
A2_6"	Nozzle	1203.20	1203.20	1.000
P1_2"	Nozzle	1203.20	1203.20	1.000
P2_2"	Nozzle	1203.20	1203.20	1.000
A3_2"	Nozzle	1203.20	1203.20	1.000
L3_2"	Nozzle	1203.20	1203.20	1.000
L4_2"	Nozzle	1203.20	1203.20	1.000
S1_3"	Nozzle	1203.20	1203.20	1.000
V1_3"	Nozzle	1203.20	1203.20	1.000
T1_2"	Nozzle	1203.20	1203.20	1.000
L1_2"	Nozzle	1203.20	1203.20	1.000
L2_2"	Nozzle	1203.20	1203.20	1.000
B2_6"	Nozzle	1203.20	1203.20	1.000
B1_3"	Nozzle	1203.20	1203.20	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Head-1	1407.16	1407.16	1.000
Shell-1	1407.13	1407.13	1.000
Shell-2	1407.13	1407.13	1.000
Shell-3	1407.13	1407.13	1.000
Head-2	1407.13	1407.13	1.000

Minimum 1.000



Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
M1_24"	450.34	1881.28	0.239
U1_8"	346.26	1881.28	0.184
A1_6"	306.21	1881.28	0.163
A2_6"	306.21	1881.28	0.163
P1_2"	129.80	1881.28	0.069
P2_2"	129.80	1881.28	0.069
A3_2"	129.80	1881.28	0.069
L3_2"	129.80	1881.28	0.069
L4_2"	129.80	1881.28	0.069
S1_3"	152.54	1881.28	0.081
V1_3"	152.54	1881.28	0.081
T1_2"	129.80	1881.28	0.069
L1_2"	129.80	1881.28	0.069
L2_2"	129.80	1881.28	0.069
B2_6"	306.21	1881.28	0.163
B1_3"	152.54	1881.28	0.081

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	2077	24.5	0.0016383	1009.79
20	30	853.75	2069	20.5	0.00029121	296.877
30	40	853.75	2069	20.5	0.00029121	296.877
40	50	853.75	2069	20.5	0.00029121	296.877
50	60	No Calc	2069	20.5	0.0013761	963.706

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	26	No Calc	...	13.2348
20	30	22	No Calc	...	3.92201
30	40	22	No Calc	...	3.92201
40	50	22	No Calc	...	3.92201
50	60	22	No Calc	...	10.6095

Minimum

4

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	853.75	No Calc	No Calc	No Calc
30	40	853.75	No Calc	No Calc	No Calc
40	50	853.75	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	1336.04	1247994	1273.42	1253310	93.523
20	30	3289.37	9661893	3067.35	9690540	230.256
30	40	3289.37	9661893	3067.35	9690540	230.256
40	50	2302.55	6763325	2147.14	6783378	161.179
50	60	1079.4	1247994	1017.13	1253310	75.5581
Total		11296	28583100.00	10572	28671080.00	790

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset, Dtl. Cent. cm	Description
10	Liqd	673.583	-16.875	16.25	...	Liquid: 10
10	Nozl	915.073	-50.625	...	...	M1_24"
10	Nozl	85.2477	-31.0305	-80	...	U1_8"
20	Sadl	332.446	95	126.525	...	Left Saddle
20	Plat	3191.1	400	111.07	...	PLAT:[1 OF 1]
20	Liqd	5020.3	150	16.25	...	Liquid 20
20	Nozl	59.1458	45	108.704	...	A1_6"
20	Nozl	59.1458	135	108.704	...	A2_6"
20	Nozl	12.744	200	103.502	...	P1_2"
20	Nozl	12.2795	250	103.502	...	P2_2"
20	Nozl	10.268	200	-103.502	...	A3_2"
20	Nozl	10.268	150	50.025	90.6096	L3_2"
20	Nozl	10.268	150	-50.025	90.6096	L4_2"
20	Wght	420	50	...	...	Lifting lug
20	Wght	200	100	...	...	Half_openpipe_A
30	Liqd	5020.3	150	16.25	...	Liquid 30
30	Nozl	20.9443	150	104.722	...	S1_3"
30	Nozl	22.3047	200	104.722	...	V1_3"
30	Nozl	13.0786	100	-103.502	...	T1_2"
30	Nozl	10.268	150	50.025	90.6096	L1_2"
30	Nozl	10.268	150	-50.025	90.6096	L2_2"
30	Wght	81.9986	...	...	...	Clip_wt
40	Sadl	332.446	115	126.525	...	New Saddle
40	Liqd	3514.21	105	16.25	...	Liquid 40
40	Nozl	62.1418	150	-108.704	...	B2_6"
40	Nozl	12.744	150	104.722	...	B1_3"
50	Liqd	673.583	21.875	16.25	...	Liquid 50

Total Weight of Each Detail Type:

Saddles	664.9
Platforms	3191.1
Liquid	14902.0
Nozzles	1326.2
Weights	702.0

Sum of the Detail Weights 20786.2 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	12087.5	12087.5	12087.5	12087.5	12087.5	12087.5
Saddles	664.9	664.9	664.9	664.9	664.9	664.9
Nozzles	1326.2	1326.2	1326.2	1326.2	1326.2	1326.2
Wld Weights	502.0	502.0	502.0	502.0	502.0	502.0
Platforms	...	...	3191.1	3191.1	3191.1	3191.1
Empty Weights	...	...	...	200.0	200.0	...

Ope Weights	...	...	...	...	...	200.0
Ope. Liquid	...	...	...	...	...	14902.0
Test Liquid	...	28570.2	...	...	...	...
Totals	14580.6	43150.8	17771.7	17971.7	17971.7	32873.7

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	14580.6 kg
Shop Test Wt.	- Fabricated weight + water (full)	43150.8 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	17771.7 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	17971.7 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	17971.7 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	32873.7 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	32098.6 kg
Field Test Wt.	- Empty weight + test medium (full)	46341.9 kg

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	50585.7	47631
20	30	194999	190852
30	40	194999	190852
40	50	136499	133596
50	60	50024.9	47631

Total	627107.0	610561.8 cm ²
Total	675.0	657.2 ft ²

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Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
M1_24"	40.49	43.44	85	300	GR 1.4	...	...	...	...
U1_8"	40.49	43.44	85	300	GR 1.4	...	...	...	...
A1_6"	81.0	86.9	85	600	GR 1.4	38.0	81.0	81.0	81.0
A2_6"	81.0	86.9	85	600	GR 1.4	38.0	81.0	81.0	81.0
P1_2"	40.49	43.44	85	300	GR 1.4	...	...	...	...
P2_2"	40.49	43.44	85	300	GR 1.4	...	...	...	...
A3_2"	81.0	86.9	85	600	GR 1.4	64.6	56.9	81.0	56.9
L3_2"	40.49	43.44	85	300	GR 1.4	...	...	...	...
L4_2"	40.49	43.44	85	300	GR 1.4	...	...	...	...
S1_3"	81.0	86.9	85	600	GR 1.4	49.7	71.7	81.0	71.7
V1_3"	81.0	86.9	85	600	GR 1.4	49.7	71.7	81.0	71.7
T1_2"	80.99	86.88	85	600	GR 1.4	...	...	...	...
L1_2"	40.49	43.44	85	300	GR 1.4	...	...	...	...
L2_2"	40.49	43.44	85	300	GR 1.4	...	...	...	...
B2_6"	48.6	52.0	85	300	GR 1.1	35.8	37.1	48.6	37.1
B1_3"	81.0	86.9	85	600	GR 1.4	49.7	71.7	81.0	71.7
Min Rating	37.079	43.441, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm ²	Maximum Pressure Pmax kgf/cm ²	Pass Fail
A1_6"	192.151	0.50	482	979	28.0	81.0	Pass
A2_6"	192.151	0.50	482	979	28.0	81.0	Pass
A3_2"	77.851	0.50	54	326	28.0	56.9	Pass
S1_3"	111.125	0.50	120	489	28.0	71.7	Pass
V1_3"	111.125	0.50	120	489	28.0	71.7	Pass
B2_6"	196.088	0.50	482	979	28.0	37.1	Pass
B1_3"	111.125	0.50	120	489	28.0	71.7	Pass

Evaluating Flange on Nozzle [A16"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(482) + 4 \cdot \text{abs}(979) \cdot 192.151 \leq \pi \cdot 192.151^3 \cdot ((80.989 - 28.008) + .500 \cdot 80.989)$$

$$= 0.84591E+04 \leq 0.20834E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A16" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(80.989, 0.500 \cdot 80.989 + 80.989 - 8459.1 / (\pi \cdot 192.151^3))$$

$$= \min(80.989, 83.530)$$

$$= 80.989 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [A26"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(482) + 4 \cdot \text{abs}(979) \cdot 192.151 \leq \pi \cdot 192.151^3 \cdot ((80.989 - 28.008) + .500 \cdot 80.989)$$

$$= 0.84591E+04 \leq 0.20834E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A26" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(80.989, 0.500 \cdot 80.989 + 80.989 - 8459.1 / (\pi \cdot 192.151^3))$$



$$= \min(80.989, 83.530)$$

$$= 80.989 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [A32"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(54) + 4 \cdot \text{abs}(326) 77.851 \leq \pi \cdot 77.851^3 [(80.989 - 28.008) + .500 \cdot 80.989]$$

$$= 0.95791E+03 \leq 0.13856E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A32" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(80.989, 0.500 \cdot 80.989 + 80.989 - 957.9 / (\pi \cdot 77.851^3))$$

$$= \min(80.989, 56.861)$$

$$= 56.861 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [S13"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(120) + 4 \cdot \text{abs}(489) 111.125 \leq \pi \cdot 111.125^3 [(80.989 - 28.008) + .500 \cdot 80.989]$$

$$= 0.21442E+04 \leq 0.40298E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle S13" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(80.989, 0.500 \cdot 80.989 + 80.989 - 2144.2 / (\pi \cdot 111.125^3))$$

$$= \min(80.989, 71.746)$$

$$= 71.746 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [V13"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(120) + 4 \cdot \text{abs}(489) 111.125 \leq \pi \cdot 111.125^3 [(80.989 - 28.008) + .500 \cdot 80.989]$$

$$= 0.21442E+04 \leq 0.40298E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle V13" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(80.989, 0.500 \cdot 80.989 + 80.989 - 2144.2 / (\pi \cdot 111.125^3))$$

$$= \min(80.989, 71.746)$$

$$= 71.746 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [B26"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(482) + 4 \cdot \text{abs}(979) 196.088 \leq \pi \cdot 196.088^3 [(48.571 - 28.008) + .500 \cdot 48.571]$$

$$= 0.84745E+04 \leq 0.10623E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle B26" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(48.571, 0.500 \cdot 48.571 + 48.571 - 8474.5 / (\pi \cdot 196.088^3))$$

$$= \min(48.571, 37.079)$$

$$= 37.079 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [B13"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(120) + 4 \cdot \text{abs}(489) 111.125 \leq \pi \cdot 111.125^3 [(80.989 - 28.008) + .500 \cdot 80.989]$$

$$= 0.21442E+04 \leq 0.40298E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle B13" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(80.989, 0.500 \cdot 80.989 + 80.989 - 2144.2 / (\pi \cdot 111.125^3))$$

$$= \min(80.989, 71.746)$$

$$= 71.746 \text{ kgf/cm}^2$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	Moderately Smooth	
Base Elevation	460	cm
Percent Wind for Hydrotest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment H or Hh	0	cm
Distance Upwind of Crest Lh	0	cm
Distance from Crest to the Vessel x	0	cm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	cm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.186 \cdot 0.961) / (1 + 1.7 \cdot 3.4 \cdot 0.186)))$
 $= \min(0.85, 0.906)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.555
Structure Height to Diameter ratio	4.274

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5: Z_g = 58978.801 \text{ cm}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 155.0 + 460.0 = 615.0 \text{ cm}$
 $= 20.177 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[K_z]$:
 Because z (20.177 ft.) $> 15 \text{ ft.}$
 $= 2.41 (z / Z_g)^{2/\alpha}$
 $= 2.41 (20.177 / 1935.0)^{2/11.5}$
 $= 1.09$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1, \text{ per Table 26.6-1}$

As there is No Hill Present: $[K_{zt}]$:
 $K_1 = 0, K_2 = 0, K_3 = 0$



Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.09 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 34.9 \text{ psf } [170.412] \text{ kgf/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 34.903 \cdot 0.85 \cdot 0.555 \cdot 10.687$$

$$= 175.8 \text{ lbs. } [79.8] \text{ kgf}$$

Element	Hgt (z) cm	K1	K2	K3	Kz	Kzt	qz kgf/m ²
Head-1	615.0	0.000	0.000	0.000	1.090	1.000	170.412
Shell-1	615.0	0.000	0.000	0.000	1.090	1.000	170.412
Shell-2	615.0	0.000	0.000	0.000	1.090	1.000	170.412
Shell-3	615.0	0.000	0.000	0.000	1.090	1.000	170.412
Head-2	615.0	0.000	0.000	0.000	1.090	1.000	170.412

Platform Load Calculations

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf	Cf
PLAT:[1 OF 1]	188192.00	615.00	170.41	3207.01	2.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf
Lifting lug	0.00	615.00	170.41	0.00
Halfopenpipe	0.00	615.00	170.41	0.00
Clipwt	0.00	615.00	170.41	0.00

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	615	249.24	9928.65	170.412	87.7313
20	30	615	248.28	74484	170.412	4185.86
30	40	615	248.28	74484	170.412	658.154
40	50	615	248.28	52138.8	170.412	460.708
50	60	615	248.28	9825.4	170.412	86.819

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h
Amplification Factor	Ap
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR]
Component strength factor	[Rpo]
Overstrength factor	[Omega0]
Response Acceleration Factor	[SMS]
Response Acceleration Factor	[SM1]
Building Importance Factor	[Ie]
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(8.40720.75)$
 = 0.099 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.099$, 0.03) per 12.8.2
 = 0.030

Loads on Building/Structure Supported Equipment:

Factor [a1]:
 = $\min(1/T_a, 2.5)$
 = $\min(1/0.03, 2.5)$
 = 2.500

Factor [a2]:
 = $\max(0, 1 - (0.4/T_a)^2)$
 = $\max(0, 1 - (0.4/0.03)^2)$
 = 0.000

Factor for force amplification factor [Hf]:
 = $1 + (a1 \cdot z/h) + (a2(z/h)^{10})$
 = $1 + (2.5 \cdot 0.5) + (0.0(0.5)^{10})$
 = 2.250

Structure ductility reduction factor [Rp]:
 = $\max(1.3, \sqrt{(1.1 \cdot R / (I_e \cdot \omega_O))})$
 = $\max(1.3, \sqrt{(1.1 \cdot 3.0 / (1.0 \cdot 2.0))})$
 = 1.300

Seismic Force per equation 13.3-1, [Fp]:



$$\begin{aligned}
 &= 0.4 \cdot \text{SDS} \cdot I_p \cdot W_p (H_f / R_\mu) (CAR / RPO) \\
 &= 0.4 \cdot 0.217 \cdot 1.5 \cdot 69300 (2.25/1.3) (1.0/1.5) \\
 &= 4729.566 \text{ kgf}
 \end{aligned}$$

Maximum value of F_p per equation 13.3-2:

$$\begin{aligned}
 &= 1.6 \cdot S_{ds} \cdot I \cdot W \\
 &= 1.6 \cdot 0.217 \cdot 1.5 \cdot 31434 = 16395.83 \text{ kgf}
 \end{aligned}$$

Minimum value of F_p per equation 13.3-3:

$$\begin{aligned}
 &= 0.3 \cdot 0.22 \cdot 1.5 \cdot 31434 \\
 &= 3074.218 \text{ kgf}
 \end{aligned}$$

Total Base Shear $V = F_p, [V]$:

$$= 4729.566 \text{ kgf}$$

Vertical load per 12.4-4, $[E_v]$:

$$\begin{aligned}
 &= 0.2 \cdot S_{ds} \cdot W \\
 &= 0.2 \cdot 0.217 \cdot 31434 \\
 &= 1366.3 \text{ kgf}
 \end{aligned}$$

Earthquake Load Calculation:

From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf
10	20	101.25	4490.53	675.652
20	30	101.25	4490.53	675.652
20	30	101.25	4490.53	675.652
30	40	101.25	4490.53	675.652
40	0	101.25	4490.53	675.652
40	50	101.25	4490.53	675.652
50	60	101.25	4490.53	675.652

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	410.000 cm
Center of Gravity of the Platforms	405.000 cm
Center of Gravity of the Liquid	410.000 cm
Center of Gravity of the Nozzles	49.217 cm
Center of Gravity of the Added Weights (Operating)	98.447 cm
Center of Gravity of the Added Weights (Empty)	98.447 cm
Center of Gravity of Bare Shell New and Cold	400.150 cm
Center of Gravity of Bare Shell Corroded	399.488 cm
Vessel CG in the Operating Condition	384.078 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	363.694 cm
Vessel CG in the Test Condition	392.123 cm

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (230.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

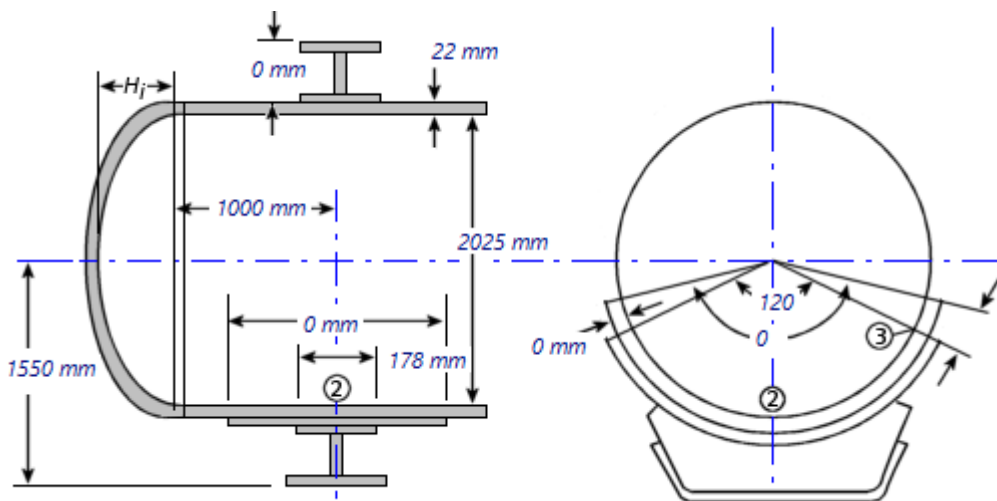
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(178.0 + 1.56 \cdot \sqrt{1024.2501 \cdot 20.5}, 2 \cdot 1000.0)$$

$$= 404.0500 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	R_m	1024.25	mm
Shell Thickness used in this Case	t	22.000	mm
Stiffened Vessel Length per 4.15.6	L	820.00	cm
Distance from Saddle to Vessel tangent	a_1 or a	1000.00	mm
Saddle Width	b_1 or b	178.00	mm
Saddle Bearing Angle	δ or θ	120.00	degrees
Inside or Mean Depth of Head	H_i or h_m	52.00	cm
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.16	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1550.00	mm
Coefficient of Friction	μ	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		24790.06	kgf
Pressure used in Analysis	P	25.532	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		603.56	1407.13
Long. Stress at Bottom of Midspan		672.10	1407.13
Long. Stress at Top of Saddles		688.29	1407.13
Long. Stress at Bottom of Saddles		609.86	1407.13
Tangential Shear in Shell		96.36	1125.71
Circ. Stress at Horn of Saddle		525.86	1758.92

Circ. Compressive Stress in Shell	22.75	1407.13
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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(5479.3/2 + 0) \cdot 1550.0/1809.1272$
 $= 7041.7\ kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(1773.57, 0) \cdot 1550.0/6200.0$
 $= 443.4\ kgf$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(4729.57, 0) \cdot 1550.0/6200.0$
 $= 1182.4\ kgf$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(4730/2 + 0) \cdot 1550.0/1809.1272$
 $= 6078.2\ kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 17748 + \max(443, 7042, 1182, 6078)$
 $= 24790.1\ kgf$

Longitudinal Wind Force [FI]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1671.158(4.62 + \pi/4(2.069 \cdot 1.2)^2)$
 $= 17392.824\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	25122.51	kgf
Transverse Shear Load	2739.64	kgf
Longitudinal Shear Load	4729.57	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness)\sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(2025.0/2 + 22.0 + 10.0)\sin(120.0/2), 2 \cdot 1024.25)$
 $= 1809.127\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0510	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm^2)/(3L))]$
 $= -24790 \cdot 100.0 [1 - (1 - 100.0/820.0 + (102.425^2 - 52.0^2) / (2 \cdot 100.0 \cdot 820.0)) / (1 + (4 \cdot 52.0^2)/(3 \cdot 820.0))]$
 $= -3634.9\ kgf \cdot m.$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 \cdot ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$
 $= 24790 \cdot 820/4 \cdot ((1 + 2(102^2 - 52^2)/(820^2)) / (1 + (4 \cdot 52)/(3 \cdot 820)) - 4 \cdot 100/820)$
 $= 23152.9\ kgf \cdot m.$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$
 $= 25.532 \cdot 1024.25/(2 \cdot 20.5) - 23152.9/(\pi \cdot 1024.3^2 \cdot 20.5)$
 $= 603.56\ kgf/cm^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 25.532 \cdot 1024.25 / (2 \cdot 20.5) + 23152.9 / (\pi \cdot 1024.3^2 \cdot 20.5)$$

$$= 672.10 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 25.532 \cdot 1024.25 / (2 \cdot 20.5) - 3634.9 / (0.1066 \cdot \pi \cdot 1024.3^2 \cdot 20.5)$$

$$= 688.29 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 25.532 \cdot 1024.25 / (2 \cdot 20.5) + 3634.9 / (0.1923 \cdot \pi \cdot 1024.3^2 \cdot 20.5)$$

$$= 609.86 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 24790(820.0 - 2 \cdot 100.0) / (820.0 + (4 \cdot 52.0 / 3))$$

$$= 17282.4 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 17282.43 / (1024.2501 \cdot 20.5)$$

$$= 96.36 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1024.25 \cdot 20.5}$$

$$= 113.025 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 24790 \cdot 0.1 / (20.5(178.0 + 113.02 + 113.02))$$

$$= -22.75 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -24790 / (4 \cdot 20.5(178.0 + 113.025 + 113.025)) - 3 \cdot 0.051 \cdot 24790 / (2 \cdot 20.5^2)$$

$$= -525.86 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(178.0 + 1.56 \cdot \sqrt{1024.25 \cdot 20.5}, 2 \cdot 100.0)$$

$$= 404.05 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 6200.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 6200.0(85.0 - 21.1)$$

$$= 4.755 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1880.0000	mm
Baseplate Thickness	Bpthk	14.0000	mm
Baseplate Width	Bpwid	204.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2110.7	kgf/cm ²
Height of Web at Center	Hw,c	480.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1014.0 \cdot 20.5)} + 230.0)^3) / 12$$

$$= 32.660 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 32.66 + 93.258 (166.923 - 10.25)^2$$

$$= 22924.107 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	454.9	20.5	10.2	93.3	95589.2	0.229E+05
Wearplate	230.0	10.0	25.5	23.0	58650.0	0.460E+04
Web	14.0	491.5	276.3	68.8	1900876.5	0.221E+05
BasePlate	204.0	14.0	529.0	28.6	1510824.1	0.374E+05
Totals	...	...	...	213.6	3565940.0	0.870E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3565940 / 213.628$$

$$= 166.923 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 24790.061$$

$$= 5045.3149 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 41.9151 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1266.4208 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1550.0 - 1014.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 711.429 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 3589.3865 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 68.8288 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1407.1342 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(24790 + 332) \cdot 204.0) / (4 \cdot 1880.0 \cdot 1407.134)}$$

$$= 12.054 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1012.5 + 22.0 + 10.0)$$

$$= 1809.127 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1809.127 / (3 - 1)$$

$$= 904.563 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot Bpwid / 2 \\
 &= 904.5635 \cdot 204.0 / 2 \\
 &= 922.655 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth) \\
 &= 25122.506 / (1880.0 \cdot 204.0) \\
 &= 6.551 \text{ kgf/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap \cdot Bp \\
 &= 922.7 \cdot 6.55 \\
 &= 6043.857 \text{ kgf}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= Rib \text{ Area} + Web \text{ Area} \\
 &= 22.96 + 63.319 \\
 &= 86.279 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P / Ar \\
 &= 6043.9 / 86.2794 \\
 &= 70.050 \text{ kgf/cm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	178.0	...	24.9	...	658.

Rib dimension [D]:

$$\begin{aligned}
 &= Saddle \text{ Width} - Web \text{ Thickness} \\
 &= 178.0 - 14.0 \\
 &= 164.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0.0 / 86.279 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= Saddle \text{ Width} / 2 \\
 &= 178.0 / 2 \\
 &= 89.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) } \\
 &= \sqrt{ (658.0 / 86.279) } \\
 &= 27.615 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk \\
 &= 1550.0 - \cos(120.0 / 2) (1012.5 + 22.0 + 10.0) - 14.0 \\
 &= 1013.750 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) } \\
 &= \sqrt{ (2 \cdot \pi^2 \cdot 2067020 / 2110.7) } \\
 &= 139.035
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL / r \\
 &= 1 \cdot 1013.75 / 27.615 \\
 &= 36.710
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot Bplen) \cdot e \cdot L / 2 \\
 &= 4729.6 / (2 \cdot 1880.0) \cdot 904.563 \cdot 1013.75 / 2
 \end{aligned}$$

= 576.732 kgf-m.

Compressive Allowable, $KL/r < Cc$ (36.7097 < 139.0349) per AISC E2-1 [Sca]:

$$= \left(1 - \frac{(KL/r)^2}{(2 \cdot Cc^2)} \right) F_y / \left(\frac{5}{3} + 3 \cdot \frac{(KL/r)}{(8 \cdot Cc)} - \frac{(KL/r)^3}{(8 \cdot Cc^3)} \right)$$

$$= \left(1 - \frac{(36.71)^2}{(2 \cdot 139.03^2)} \right) 2111 / \left(\frac{5}{3} + 3 \cdot \frac{(36.71)}{(8 \cdot 139.03)} - \frac{(36.71^3)}{(8 \cdot 139.03^3)} \right)$$

$$= 1155 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 70.05/1155.24 + (576.73 \cdot 89.0/6579711) / 1407.13$$

$$= 0.615$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	164.0	0.0	23.0	0.0	658.
Web	904.6	14.0	0.0	126.6	0.0	20.7
Totals	...	...	...	149.6	...	678.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/149.599$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178.0/2$$

$$= 89.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1550.0 - \cos(1044.5/2) (14.0 + 0.0 + 0.0) - 0.0$$

$$= 491.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 678.3/149.599 }$$

$$= 21.294 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 491.5/21.294$$

$$= 23.082$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 4729.566 / (2 \cdot 1880.0)$$

$$= 1.258 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.258 \cdot 904.563 \cdot 491.5$$

$$= 559.238 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 559.238/76217.445$$

$$= 733.740 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < Cc$ (23.0816 < 139.0349) per AISC E2-1 [Sca]:

$$= \left(1 - \frac{(KL/r)^2}{(2 \cdot Cc^2)} \right) F_y / \left(\frac{5}{3} + 3 \cdot \frac{(KL/r)}{(8 \cdot Cc)} - \frac{(KL/r)^3}{(8 \cdot Cc^3)} \right)$$

$$= \left(1 - \frac{(23.08)^2}{(2 \cdot 139.03^2)} \right) 2111 / \left(\frac{5}{3} + 3 \cdot \frac{(23.08)}{(8 \cdot 139.03)} - \frac{(23.08^3)}{(8 \cdot 139.03^3)} \right)$$

$$= 1204 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$



$$= 80.76/1204.4 + (559.24 \cdot 89.0/678.335)/1407.13$$

$$= 0.588$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	1529.52 kgf/cm ²
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1101.23 kgf/cm ²
Area Available in a Single Bolt	BlArea	2.9291 cm ²
Saddle Load QO (Weight)	QO	18080.8 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1182.4 kgf
Maximum Transverse Force	Ft	2739.6 kgf
Maximum Longitudinal Force	Fl	4729.6 kgf
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 4729.6 / (2.929 \cdot 4)$$

$$= 403.7 \text{ kgf/cm}^2. \text{ Must be less than } 1019.7 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2739.6 / (2.929 \cdot 4)$$

$$= 233.8 \text{ kgf/cm}^2. \text{ Must be less than } 1019.7 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 4729.57 / (1019.68 \cdot 4.0)$$

$$= 1.1596 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1550.0 \cdot 2739.64 + 0.0$$

$$= 4246.44 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 4246.44 / 18080.81$$

$$= 234.86 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 1.1596, 0.0]$$

$$= 1.1596 \text{ cm}^2$$

Note: Saddle and wear plate material are both specified as impact tested.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (230.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56\sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(178.0 + 1.56\sqrt{(1024.2501 \cdot 20.5)}, 2 \cdot 1000.0)$$

$$= 404.0500 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1024.25	mm
Shell Thickness used in this Case	t	22.000	mm
Stiffened Vessel Length per 4.15.6	L	820.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	178.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	51.80	cm
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1550.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		22093.36	kgf
Pressure used in Analysis	P	25.532	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	607.27	1407.13
Long. Stress at Bottom of Midspan	668.39	1407.13
Long. Stress at Top of Saddles	682.69	1407.13
Long. Stress at Bottom of Saddles	612.97	1407.13
Tangential Shear in Shell	85.90	1125.71
Circ. Stress at Horn of Saddle	468.66	1758.92
Circ. Compressive Stress in Shell	20.28	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(5479.3/2 + 0) \cdot 1550.0/1809.1272$$

$$= 7041.7 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(1773.57, 0) \cdot 1550.0/6200.0$$

$$= 443.4 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(4729.57, 0) \cdot 1550.0/6200.0$$

$$= 1182.4 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(4730/2 + 0) \cdot 1550.0/1809.1272$$

$$= 6078.2 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 15052 + \max(443, 7042, 1182, 6078)$$

$$= 22093.4 \text{ kgf}$$

Longitudinal Wind Force [F_L]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1671.158 (4.62 + \pi/4 (2.069 \cdot 1.2)^2)$
 $= 17392.824 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	22425.80	kgf
Transverse Shear Load	2739.64	kgf
Longitudinal Shear Load	4729.57	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(2025.0/2 + 22.0 + 10.0) \sin(120.0/2), 2 \cdot 1024.25)$
 $= 1809.127 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0510	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -22093 \cdot 100.0 [1 - (1 - 100.0/820.0 + (102.425^2 - 51.8^2) / (2 \cdot 100.0 \cdot 820.0)) / (1 + (4 \cdot 51.8) / (3 \cdot 820.0))]$
 $= -3231.2 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 22093 \cdot 820 / 4 ((1 + 2(102^2 - 52^2)/(820^2)) / (1 + (4 \cdot 52) / (3 \cdot 820)) - 4 \cdot 100/820)$
 $= 20649.7 \text{ kgf-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t)$
 $= 25.532 \cdot 1024.25 / (2 \cdot 20.5) - 20649.7 / (\pi \cdot 1024.3^2 \cdot 20.5)$
 $= 607.27 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$
 $= 25.532 \cdot 1024.25 / (2 \cdot 20.5) + 20649.7 / (\pi \cdot 1024.3^2 \cdot 20.5)$
 $= 668.39 \text{ kgf/cm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 $= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$
 $= 25.532 \cdot 1024.25 / (2 \cdot 20.5) - 3231.2 / (0.1066 \cdot \pi \cdot 1024.3^2 \cdot 20.5)$
 $= 682.69 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:
 $= P \cdot R_m / (2t) + M1 / (K1^* \cdot \pi \cdot R_m^2 \cdot t)$
 $= 25.532 \cdot 1024.25 / (2 \cdot 20.5) + 3231.2 / (0.1923 \cdot \pi \cdot 1024.3^2 \cdot 20.5)$
 $= 612.97 \text{ kgf/cm}^2$

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L - 2a) / (L + (4 \cdot h_m/3))$
 $= 22093 (820.0 - 2 \cdot 100.0) / (820.0 + (4 \cdot 51.8/3))$
 $= 15407.0 \text{ kgf}$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:
 $= K2 \cdot T / (R_m \cdot t)$
 $= 1.1707 \cdot 15407.04 / (1024.2501 \cdot 20.5)$
 $= 85.90 \text{ kgf/cm}^2$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{ R_m \cdot t }$
 $= 0.78 \cdot \sqrt{ 1024.25 \cdot 20.5 }$

= 113.025 mm

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:
 = $-K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$
 = $-0.7603 \cdot 22093 \cdot 0.1 / (20.5(178.0 + 113.02 + 113.02))$
 = -20.28 kgf/cm^2

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:
 = $-Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$
 = $-22093 / (4 \cdot 20.5(178.0 + 113.025 + 113.025)) - 3 \cdot 0.051 \cdot 22093 / (2 \cdot 20.5^2)$
 = -468.66 kgf/cm^2

Effective reinforcing plate width (4.15.24) [B1]:
 = $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 = $\min(178.0 + 1.56 \cdot \sqrt{1024.25 \cdot 20.5}, 2 \cdot 100.0)$
 = 404.05 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1880.0000	mm
Baseplate Thickness	Bpthk	14.0000	mm
Baseplate Width	Bpwid	204.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2110.7	kgf/cm ²
Height of Web at Center	Hw,c	480.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 = $((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})^3) / 12$
 = $((1.56 \cdot \sqrt{1014.0 \cdot 20.5} + 230.0)^3) / 12$
 = 32.660 cm⁴

Resolved Inertia for the Shell [I,shell]:
 = ShellInertia + AShell(C1 - Y)²
 = 32.66 + 93.258(166.923 - 10.25)²
 = 22924.107 cm⁴

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	454.9	20.5	10.2	93.3	95589.2	0.229E+05
Wearplate	230.0	10.0	25.5	23.0	58650.0	0.460E+04
Web	14.0	491.5	276.3	68.8	1900876.5	0.221E+05
BasePlate	204.0	14.0	529.0	28.6	1510824.1	0.374E+05
Totals	...	...	...	213.6	3565940.0	0.870E+05

Distance to Centroid [C1]:
 = AY / A
 = 3565940/213.628
 = 166.923 mm

Angle [β]:
 = 180 - Saddle Angle/2
 = 180 - 120.0/2
 = 120.0

Saddle Splitting Coefficient [K1]:
 = $(1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$
 = $(1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$
 = 0.2035

Saddle Splitting Force [Fh]:
 = K1 • Q

$$= 0.204 \cdot 22093.357$$

$$= 4496.4775 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 37.3555 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1266.4208 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1550.0 - 1014.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 711.429 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 3198.9270 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 61.3415 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1407.1342 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(22093 + 332)204.0 / (4 \cdot 1880.0 \cdot 1407.134)}$$

$$= 11.389 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1012.5 + 22.0 + 10.0)$$

$$= 1809.127 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1809.127 / (3 - 1)$$

$$= 904.563 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 904.5635 \cdot 204.0 / 2$$

$$= 922.655 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 22425.805 / (1880.0 \cdot 204.0)$$

$$= 5.847 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 922.7 \cdot 5.85$$

$$= 5395.097 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 22.96 + 63.319$$

$$= 86.279 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 5395.1/86.2794$$

$$= 62.531 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	178.0	...	24.9	...	658.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 178.0 - 14.0$$

$$= 164.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/86.279$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178.0/2$$

$$= 89.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(658.0/86.279)}$$

$$= 27.615 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1550.0 - \cos(120.0/2) (1012.5 + 22.0 + 10.0) - 14.0$$

$$= 1013.750 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2067020/2110.7)}$$

$$= 139.035$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1013.75/27.615$$

$$= 36.710$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 4729.6 / (2 \cdot 1880.0) \cdot 904.563 \cdot 1013.75/2$$

$$= 576.732 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($36.7097 < 139.0349$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (36.71)^2 / (2 \cdot 139.03^2)) 2111 /$$

$$(5/3 + 3 \cdot (36.71) / (8 \cdot 139.03) - (36.71^3) / (8 \cdot 139.03^3))$$

$$= 1155 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 62.53/1155.24 + (576.73 \cdot 89.0/6579711) / 1407.13$$

$$= 0.609$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	164.0	0.0	23.0	0.0	658.
Web	904.6	14.0	0.0	126.6	0.0	20.7
Totals	...	...	...	149.6	...	678.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/149.599$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178.0/2$$

$$= 89.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1550.0 - \cos(1044.5/2) (14.0 + 0.0 + 0.0) - 0.0$$

$$= 491.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{678.3/149.599}$$

$$= 21.294 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 491.5/21.294$$

$$= 23.082$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 4729.566 / (2 \cdot 1880.0)$$

$$= 1.258 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.258 \cdot 904.563 \cdot 491.5$$

$$= 559.238 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 559.238/76217.445$$

$$= 733.740 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($23.0816 < 139.0349$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (23.08)^2 / (2 \cdot 139.03^2)) 2111 /$$

$$(5/3 + 3 \cdot (23.08) / (8 \cdot 139.03) - (23.08)^3 / (8 \cdot 139.03^3))$$

$$= 1204 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 72.09/1204.4 + (559.24 \cdot 89.0/678.335) / 1407.13$$

$$= 0.581$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	1529.52 kgf/cm ²
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1101.23 kgf/cm ²
Area Available in a Single Bolt	BltArea	2.9291 cm ²
Saddle Load Q0 (Weight)	Q0	15384.1 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1182.4 kgf
Maximum Transverse Force	Ft	2739.6 kgf
Maximum Longitudinal Force	Fl	4729.6 kgf
Saddle Bolted to Steel Foundation		Yes

Friction force due to thermal growth, [Ff]:

$$= 5424.2 \text{ kgf}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 5424.2 / (2.929 \cdot 4)$$

$$= 463.0 \text{ kgf/cm}^2. \text{ Must be less than } 1019.7 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2739.6 / (2.929 \cdot 4)$$

$$= 233.8 \text{ kgf/cm}^2. \text{ Must be less than } 1019.7 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (BoltShearAllowable \cdot Nbolts)$
= $4729.57 / (1019.68 \cdot 4.0)$
= 1.1596 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1550.0 \cdot 2739.64 + 0.0$
= 4246.44 kgf-m.

Eccentricity (e):
= $Rmom / QO$
= $4246.44 / 15384.11$
= 276.03 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0.0, 1.1596, 0.0]$
= 1.1596 cm²

Note: Saddle and wear plate material are both specified as impact tested.

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (230.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

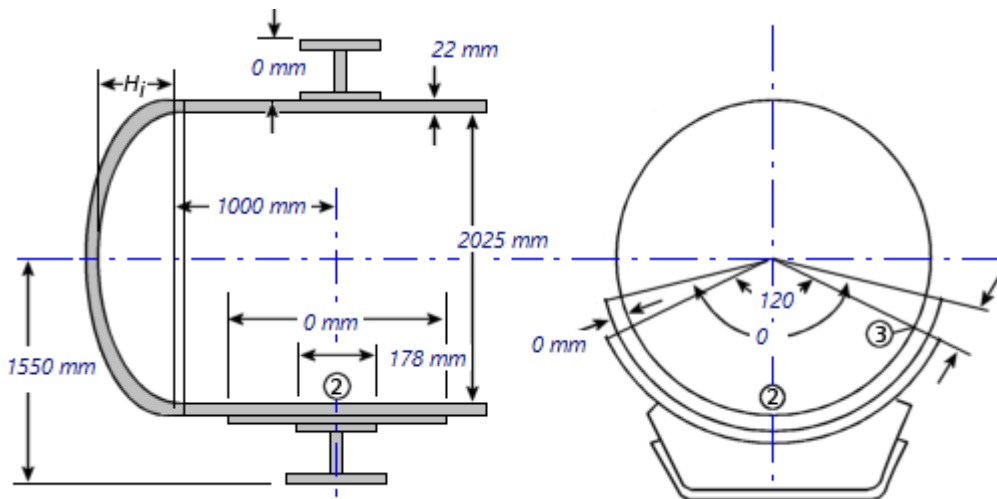
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(178.0 + 1.56 \cdot \sqrt{1023.5 \cdot 22.0}, 2 \cdot 1000.0)$$

$$= 412.0883 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1023.50	mm
Shell Thickness used in this Case	t	22.000	mm
Stiffened Vessel Length per 4.15.6	L	820.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	178.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	51.93	cm
Shell Allowable Stress used in Calculation		2386.01	kgf/cm ²
Head Allowable Stress used in Calculation		2386.01	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1550.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	P	26412.49	kgf
Pressure used in Analysis		36.513	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		815.27	2386.01
Long. Stress at Bottom of Midspan		883.42	2386.01
Long. Stress at Top of Saddles		899.50	2386.01
Long. Stress at Bottom of Saddles		821.55	2386.01
Tangential Shear in Shell		95.75	1413.05
Circ. Stress at Horn of Saddle		490.56	3579.01

Circ. Compressive Stress in Shell	22.15	2386.01
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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(1808.2/2 + 0) \cdot 1550.0/1809.1272$
 $= 2323.8\ kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind};\ Sum\ of\ X\ Forces) \cdot B / L_s$
 $= \max(585.28,\ 0) \cdot 1550.0/6200.0$
 $= 146.3\ kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(F_{wl},\ F_{wt},\ F_{sl},\ F_{st})$
 $= 24089 + \max(146,\ 2324,\ 0,\ 0)$
 $= 26412.5\ kgf$

Longitudinal Wind Force [FI]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1671.158(4.62 + \pi/4(2.069 \cdot 1.2)^2)$
 $= 17392.824\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	26744.94	kgf
Transverse Shear Load	904.08	kgf
Longitudinal Shear Load	585.28	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness)\sin(\theta/2),\ 2 \cdot R_m)$
 $= \min(2(2025.0/2 + 22.0 + 10.0)\sin(120.0/2),\ 2 \cdot 1023.5)$
 $= 1809.127\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0510	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -26412 \cdot 100.0 [1 - (1 - 100.0/820.0 + (102.35^2 - 51.925^2) / (2 \cdot 100.0 \cdot 820.0)) / (1 + (4 \cdot 51.93) / (3 \cdot 820.0))]$
 $= -3871.3\ kgf \cdot m.$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(R_m^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$
 $= 26412 \cdot 820 / 4 ((1 + 2(102^2 - 52^2)/(820^2)) / (1 + (4 \cdot 52) / (3 \cdot 820)) - 4 \cdot 100/820)$
 $= 24672.8\ kgf \cdot m.$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$
 $= 36.513 \cdot 1023.5 / (2 \cdot 22.0) - 24672.8 / (\pi \cdot 1023.5^2 \cdot 22.0)$
 $= 815.27\ kgf/cm^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$
 $= 36.513 \cdot 1023.5 / (2 \cdot 22.0) + 24672.8 / (\pi \cdot 1023.5^2 \cdot 22.0)$
 $= 883.42\ kgf/cm^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:
 $= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$
 $= 36.513 \cdot 1023.5 / (2 \cdot 22.0) - 3871.3 / (0.1066 \cdot \pi \cdot 1023.5^2 \cdot 22.0)$
 $= 899.50\ kgf/cm^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 36.513 \cdot 1023.5 / (2 \cdot 22.0) + 3871.3 / (0.1923 \cdot \pi \cdot 1023.5^2 \cdot 22.0)$$

$$= 821.55 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 26412(820.0 - 2 \cdot 100.0) / (820.0 + (4 \cdot 51.93 / 3))$$

$$= 18415.6 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 18415.58 / (1023.5 \cdot 22.0)$$

$$= 95.75 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1023.5 \cdot 22.0)}$$

$$= 117.044 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 26412 \cdot 0.1 / (22.0(178.0 + 117.04 + 117.04))$$

$$= -22.15 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -26412 / (4 \cdot 22.0(178.0 + 117.044 + 117.044)) - 3 \cdot 0.051 \cdot 26412 / (2 \cdot 22.0^2)$$

$$= -490.56 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(178.0 + 1.56 \cdot \sqrt{(1023.5 \cdot 22.0)}, 2 \cdot 100.0)$$

$$= 412.09 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1880.0000	mm
Baseplate Thickness	Bpthk	14.0000	mm
Baseplate Width	Bpwid	204.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2090.3	kgf/cm ²
Height of Web at Center	Hw,c	480.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1012.5 \cdot 22.0)} + 230.0)^3) / 12$$

$$= 41.068 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 41.068 + 101.822(162.041 - 11.0)^2$$

$$= 23269.938 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	462.8	22.0	11.0	101.8	112004.1	0.233E+05
Wearplate	230.0	10.0	27.0	23.0	62100.0	0.420E+04
Web	14.0	491.5	277.8	68.8	1911198.0	0.231E+05
BasePlate	204.0	14.0	530.5	28.6	1515108.1	0.388E+05
Totals	...	...	...	222.2	3600410.2	0.893E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3600410/222.192$$

$$= 162.041 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 26412.492$$

$$= 5375.5146 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 44.6583 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1254.1849 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1550.0 - 1012.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 712.669 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 3830.9688 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 69.5080 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1393.5388 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(26412 + 332)204.0 / (4 \cdot 1880.0 \cdot 1407.134)}$$

$$= 12.437 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1012.5 + 22.0 + 10.0)$$

$$= 1809.127 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1809.127 / (3 - 1)$$

$$= 904.563 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 904.5635 \cdot 204.0/2$$

$$= 922.655 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 26744.939 / (1880.0 \cdot 204.0)$$

$$= 6.974 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 922.7 \cdot 6.97$$

$$= 6434.175 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 22.96 + 63.319 \\
 &= 86.279 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Compressive Stress [Sc]:} \\
 &= P/Ar \\
 &= 6434.2/86.2794 \\
 &= 74.574 \text{ kgf/cm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	178.0	...	24.9	...	658.

$$\begin{aligned}
 &\text{Rib dimension [D]:} \\
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 178.0 - 14.0 \\
 &= 164.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Distance to Centroid from Datum [ytot]:} \\
 &= AY / A \\
 &= 0.0/86.279 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Distance to Centroid [C1]:} \\
 &= \text{Saddle Width} / 2 \\
 &= 178.0/2 \\
 &= 89.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Radius of Gyration [r]:} \\
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(658.0/86.279)} \\
 &= 27.615 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Length of Outer Rib [L]:} \\
 &= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1550.0 - \cos(120.0/2) (1012.5 + 22.0 + 10.0) - 14.0 \\
 &= 1013.750 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Intermediate Term [Cc]:} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 2067020/2090.3)} \\
 &= 139.711
 \end{aligned}$$

$$\begin{aligned}
 &\text{Slenderness ratio [KL/r]:} \\
 &= KL/r \\
 &= 1 \cdot 1013.75/27.615 \\
 &= 36.710
 \end{aligned}$$

$$\begin{aligned}
 &\text{Bending Moment [Rm]:} \\
 &= FI / (2 \cdot Bplen) \cdot e \cdot L / 2 \\
 &= 585.3 / (2 \cdot 1880.0) \cdot 904.563 \cdot 1013.75/2 \\
 &= 71.370 \text{ kgf-m.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Compressive Allowable, } KL/r < Cc \text{ (} 36.7097 < 139.7115 \text{) per AISC E2-1 [Sca]:} \\
 &= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3)) \\
 &= (1 - (36.71)^2 / (2 \cdot 139.71^2)) 2090 / \\
 &\quad (5/3 + 3 \cdot (36.71) / (8 \cdot 139.71) - (36.71^3) / (8 \cdot 139.71^3)) \\
 &= 1145 \text{ kgf/cm}^2
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= Sc/Sca + (Rm \cdot C1 / I) / Sba \\
 &= 74.57/1144.77 + (71.37 \cdot 89.0/6579711) / 1393.54 \\
 &= 0.134
 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	164.0	0.0	23.0	0.0	658.
Web	904.6	14.0	0.0	126.6	0.0	20.7
Totals	...	...	...	149.6	...	678.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/149.599$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178.0/2$$

$$= 89.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1550.0 - \cos(1044.5/2)(14.0 + 0.0 + 0.0) - 0.0$$

$$= 491.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(678.3/149.599)}$$

$$= 21.294 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 491.5/21.294$$

$$= 23.082$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 585.278 / (2 \cdot 1880.0)$$

$$= 0.156 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.156 \cdot 904.563 \cdot 491.5$$

$$= 69.205 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 69.205/76217.445$$

$$= 90.799 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($23.0816 < 139.7115$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (23.08)^2 / (2 \cdot 139.71^2)) 2090 /$$

$$(5/3 + 3 \cdot (23.08) / (8 \cdot 139.71) - (23.08)^3 / (8 \cdot 139.71^3))$$

$$= 1193 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 85.97/1193.12 + (69.21 \cdot 89.0/678.335) / 1393.54$$

$$= 0.137$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification	SA-36	
Bolt Allowable Stress	Stba	1529.52 kgf/cm ²
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1101.24 kgf/cm ²
Area Available in a Single Bolt	BltArea	2.9291 cm ²



Saddle Load Q0 (Weight)	Q0	24421.2 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	146.3 kgf
Maximum Transverse Force	Ft	904.1 kgf
Maximum Longitudinal Force	F1	585.3 kgf
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 = $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $585.3 / (2.929 \cdot 4)$
 = 50.0 kgf/cm². Must be less than 1019.7 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 = $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $904.1 / (2.929 \cdot 4)$
 = 77.2 kgf/cm². Must be less than 1019.7 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 = $585.28 / (1019.68 \cdot 4.0)$
 = 0.1435 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + \text{Sum of X Moments}$
 = $1550.0 \cdot 904.08 + 0.0$
 = 1401.33 kgf-m.

Eccentricity (e):
 = $Rmom / Q0$
 = $1401.33 / 24421.18$
 = 57.38 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 = $\max[0.0, 0.1435, 0.0]$
 = 0.1435 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (230.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{ Rm \cdot t }, 2a)$
 = $\min(178.0 + 1.56 \cdot \sqrt{ 1023.5 \cdot 22.0 }, 2 \cdot 1000.0)$
 = 412.0883 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1023.50	mm
Shell Thickness used in this Case	t	22.000	mm
Stiffened Vessel Length per 4.15.6	L	820.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	178.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	51.72	cm

Shell Allowable Stress used in Calculation	2386.01	kgf/cm ²
Head Allowable Stress used in Calculation	2386.01	kgf/cm ²
Circumferential Efficiency in Plane of Saddle	1.00	
Circumferential Efficiency at Mid-Span	1.00	
Distance from Saddle Base to Centerline	B	1550.00 mm
Coefficient of Friction	mu	0.00

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	23912.02	kgf
Pressure used in Analysis	P	36.513 kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	818.47	2386.01
Long. Stress at Bottom of Midspan	880.22	2386.01
Long. Stress at Top of Saddles	894.64	2386.01
Long. Stress at Bottom of Saddles	824.24	2386.01
Tangential Shear in Shell	86.71	1413.05
Circ. Stress at Horn of Saddle	444.12	3579.01
Circ. Compressive Stress in Shell	20.05	2386.01

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(1808.2/2 + 0) \cdot 1550.0/1809.1272$
 $= 2323.8 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(585.28, 0) \cdot 1550.0/6200.0$
 $= 146.3 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$
 $= 21588 + \max(146, 2324, 0, 0)$
 $= 23912.0 \text{ kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1671.158(4.62 + \pi/4(2.069 \cdot 1.2)^2)$
 $= 17392.824 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	24244.47	kgf
Transverse Shear Load	904.08	kgf
Longitudinal Shear Load	585.28	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(2025.0/2 + 22.0 + 10.0) \sin(120.0/2), 2 \cdot 1023.5)$
 $= 1809.127 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0510	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -23912 \cdot 100.0 [1 - (1 - 100.0/820.0 + (102.35^2 - 51.725^2) /$

$$\frac{(2 \cdot 100.0 \cdot 820.0)}{(1 + (4 \cdot 51.72)/(3 \cdot 820.0))} \\ = -3495.9 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= \frac{Q \cdot L}{4} \left(\frac{1 + 2(Rm^2 - hm^2)/(L^2)}{1 + (4hm^2)/(3L)} - 4a/L \right) \\ = \frac{23912 \cdot 820}{4} \left(\frac{1 + 2(1023.5^2 - 52^2)/(820^2)}{1 + (4 \cdot 52)/(3 \cdot 820)} - 4 \cdot 100/820 \right) \\ = 22353.7 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t) \\ = 36.513 \cdot 1023.5/(2 \cdot 22.0) - 22353.7/(\pi \cdot 1023.5^2 \cdot 22.0) \\ = 818.47 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t) \\ = 36.513 \cdot 1023.5/(2 \cdot 22.0) + 22353.7/(\pi \cdot 1023.5^2 \cdot 22.0) \\ = 880.22 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t) \\ = 36.513 \cdot 1023.5/(2 \cdot 22.0) - 3495.9/(0.1066 \cdot \pi \cdot 1023.5^2 \cdot 22.0) \\ = 894.64 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = 36.513 \cdot 1023.5/(2 \cdot 22.0) + 3495.9/(0.1923 \cdot \pi \cdot 1023.5^2 \cdot 22.0) \\ = 824.24 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= \frac{Q(L - 2a)}{L + (4 \cdot hm/3)} \\ = \frac{23912(820.0 - 2 \cdot 100.0)}{820.0 + (4 \cdot 51.72/3)} \\ = 16677.2 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t) \\ = 1.1707 \cdot 16677.17 / (1023.5 \cdot 22.0) \\ = 86.71 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{Rm \cdot t} \\ = 0.78 \cdot \sqrt{1023.5 \cdot 22.0} \\ = 117.044 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\ = -0.7603 \cdot 23912 \cdot 0.1 / (22.0(178.0 + 117.04 + 117.04)) \\ = -20.05 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -\frac{Q}{4 \cdot t(b + X1 + X2)} - \frac{3 \cdot K7 \cdot Q}{2 \cdot t^2} \\ = -\frac{23912}{4 \cdot 22.0(178.0 + 117.044 + 117.044)} - \frac{3 \cdot 0.051 \cdot 23912}{2 \cdot 22.0^2} \\ = -444.12 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a) \\ = \min(178.0 + 1.56 \cdot \sqrt{1023.5 \cdot 22.0}, 2 \cdot 100.0) \\ = 412.09 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1880.0000	mm
Baseplate Thickness	Bpthk	14.0000	mm
Baseplate Width	Bpwid	204.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2090.3	kgf/cm ²

Height of Web at Center Hw,c 480.0 mm
 Friction Coefficient mu 0.000

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1012.5 \cdot 22.0)} + 230.0)^3) / 12$$

$$= 41.068 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 41.068 + 101.822 (162.041 - 11.0)^2$$

$$= 23269.938 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	462.8	22.0	11.0	101.8	112004.1	0.233E+05
Wearplate	230.0	10.0	27.0	23.0	62100.0	0.420E+04
Web	14.0	491.5	277.8	68.8	1911198.0	0.231E+05
BasePlate	204.0	14.0	530.5	28.6	1515108.1	0.388E+05
Totals	...	...	...	222.2	3600410.2	0.893E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3600410 / 222.192$$

$$= 162.041 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 23912.02$$

$$= 4866.6147 \text{ kgf}$$

Tension Stress, St = (Fh/As) = 40.4305 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1254.1849 kgf/cm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1550.0 - 1012.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 712.669 \text{ mm}$$

Bending Moment, M = Fh * d = 3468.2908 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 62.9277 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1393.5388 kgf/cm²

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(23912 + 332)204.0 / (4 \cdot 1880.0 \cdot 1407.134)}$$

$$= 11.841 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1012.5 + 22.0 + 10.0)$$

$$= 1809.127 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1809.127 / (3 - 1)$$

$$= 904.563 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 904.5635 \cdot 204.0 / 2$$

$$= 922.655 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 24244.467 / (1880.0 \cdot 204.0)$$

$$= 6.322 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 922.7 \cdot 6.32$$

$$= 5832.623 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 22.96 + 63.319$$

$$= 86.279 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 5832.6 / 86.2794$$

$$= 67.602 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	178.0	...	24.9	...	658.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 178.0 - 14.0$$

$$= 164.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0.0 / 86.279$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178.0 / 2$$

$$= 89.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(658.0 / 86.279)}$$

$$= 27.615 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1550.0 - \cos(120.0/2) (1012.5 + 22.0 + 10.0) - 14.0$$

$$= 1013.750 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2067020 / 2090.3)}$$

$$= 139.711$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1013.75/27.615$$

$$= 36.710$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 585.3 / (2 \cdot 1880.0) \cdot 904.563 \cdot 1013.75/2$$

$$= 71.370 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ (36.7097 < 139.7115) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (36.71)^2 / (2 \cdot 139.71^2)) 2090 /$$

$$(5/3 + 3 \cdot (36.71) / (8 \cdot 139.71) - (36.71^3) / (8 \cdot 139.71^3))$$

$$= 1145 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 67.6/1144.77 + (71.37 \cdot 89.0/6579711) / 1393.54$$

$$= 0.128$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	164.0	0.0	23.0	0.0	658.
Web	904.6	14.0	0.0	126.6	0.0	20.7
Totals	...	...	...	149.6	...	678.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/149.599$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178.0/2$$

$$= 89.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1550.0 - \cos(1044.5/2) (14.0 + 0.0 + 0.0) - 0.0$$

$$= 491.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (678.3/149.599) }$$

$$= 21.294 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 491.5/21.294$$

$$= 23.082$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 585.278 / (2 \cdot 1880.0)$$

$$= 0.156 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.156 \cdot 904.563 \cdot 491.5$$

$$= 69.205 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 69.205/76217.445$$

$$= 90.799 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (23.0816 < 139.7115) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (23.08)^2 / (2 \cdot 139.71^2)) 2090 / (5/3 + 3 \cdot (23.08) / (8 \cdot 139.71) - (23.08^3) / (8 \cdot 139.71^3))$$

$$= 1193 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 77.93 / 1193.12 + (69.21 \cdot 89.0 / 678.335) / 1393.54$$

$$= 0.130$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	1529.52 kgf/cm ²
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1101.24 kgf/cm ²
Area Available in a Single Bolt	BltArea	2.9291 cm ²
Saddle Load QO (Weight)	QO	21920.7 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	146.3 kgf
Maximum Transverse Force	Ft	904.1 kgf
Maximum Longitudinal Force	Fl	585.3 kgf
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 585.3 / (2.929 \cdot 4)$$

$$= 50.0 \text{ kgf/cm}^2. \text{ Must be less than } 1019.7 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 904.1 / (2.929 \cdot 4)$$

$$= 77.2 \text{ kgf/cm}^2. \text{ Must be less than } 1019.7 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 585.28 / (1019.68 \cdot 4.0)$$

$$= 0.1435 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1550.0 \cdot 904.08 + 0.0$$

$$= 1401.33 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 1401.33 / 21920.71$$

$$= 63.93 \text{ mm} < Bplen / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.1435, 0.0]$$

$$= 0.1435 \text{ cm}^2$$



Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: M124"

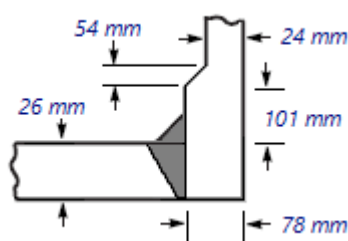
From: 10

Pressure for Reinforcement Calculations	P	28.048	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1407.16	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.16	kgf/cm ²
Inside Diameter of Elliptical Head	D	2025.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	26.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	101.0000	mm
Height of Beveled Transition	L'	54.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		78.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: M124"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 24.000 in.
 Actual Thickness Used in Calculation 0.945 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 28.05 \cdot 2028.0 \cdot 0.8987 / (2 \cdot 1407.16 \cdot 1.0 - 0.2 \cdot 28.05)$
 $= 18.2010 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 28.05 \cdot 304.8 / (1203 \cdot 1.0 + 0.4 \cdot 28.05)$
 $= 7.0397 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 54.0 / 0.5773$
 $= 93.5308 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

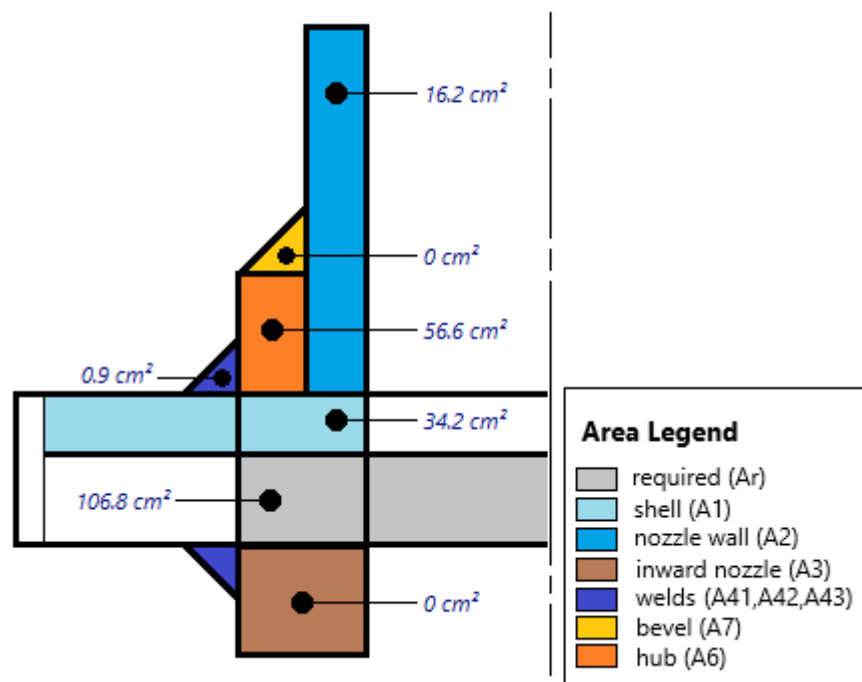
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1129.2001	mm
Parallel to Vessel Wall, opening length	d	564.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	61.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1203.2/1407.2)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1203.2/1407.2)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	106.799	NA	NA
Area in Shell	A1	34.167	NA	NA
Area in Nozzle Wall	A2	16.194	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	56.562	NA	NA
TOTAL AREA AVAILABLE	Atot	107.778	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (564.6 \cdot 18.201 \cdot 1.0 + 2 \cdot 76.5 \cdot 18.201 \cdot 1.0 \cdot (1 - 0.86)) \\
 &= 106.799 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 564.6(1.0 \cdot 24.5 - 1.0 \cdot 18.201) - 2 \cdot 76.5 \\
 &\quad (1.0 \cdot 24.5 - 1.0 \cdot 18.201) \cdot (1 - 0.855) \\
 &= 34.167 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tlnp)(tn - trn)fr2 \\
 &= (2 \cdot 61.25)(22.5 - 7.04)0.8551 \\
 &= 16.194 \text{ cm}^2
 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2 \\
 &= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551 \\
 &= 0.855 \text{ cm}^2
 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(61.3, 300.0, 101.0)) \cdot (78.0 - 24.0) \cdot 0.855$$

$$= 56.562 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 8.5397 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= t_a$
 $= 8.5397 \text{ mm}$

Available Nozzle Neck Thickness = 24.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification $-29 \text{ }^{\circ}\text{C}$

Calculated Minimum Design Metal Temperature $-104 \text{ }^{\circ}\text{C}$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification $-29 \text{ }^{\circ}\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $-29 \text{ }^{\circ}\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification $-29 \text{ }^{\circ}\text{C}$

Flange MDMT with Temp reduction per UCS-66(i)(2) $-49 \text{ }^{\circ}\text{C}$

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

Design Pressure/Ambient Rating = $28.05/43.44 = 0.646$

Weld Size Calculations, Description: M124"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (106.7994 - 34.1671 + 2 \cdot 76.5 \cdot 0.8551 \cdot (1.0 \cdot 24.5 - 18.201)) \cdot 1407)$$

$$= 113800.83 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (72.7559 + 0.8551 - 0.0 \cdot 0.86) \cdot 1407$$

$$= 103582.20 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (72.7559 + 0.0 + 0.8551 + (32.0518)) \cdot 1407$$



= 148684.17 kgf

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r 1)) \cdot S \\ &= (72.7559 + 0.0 + 0.8551 + 0.0 + (32.0518)) \cdot 1407 \\ &= 148684.17 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 1203 \\ &= 56455. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 293.55) \cdot (78.0 - 1.5) \cdot 0.7 \cdot 1203 \\ &= 594195. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 609.6 \cdot (20.0 - 0.0) \cdot 0.74 \cdot 1407 \\ &= 199420. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (56455 + 594195) = 650650 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (56455 + 0 + 199420 + 0) = 255875 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (56455 + 199420 + 0) = 255875 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 650649 kgf, must exceed W = 113800 kgf or W1 = 103582 kgf
Path 2-2 = 255874 kgf, must exceed W = 113800 kgf or W2 = 148684 kgf
Path 3-3 = 255874 kgf, must exceed W = 113800 kgf or W3 = 148684 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 28 kgf/cm²

The Drop for this Nozzle is : 25.3748 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 351.3748 mm

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Input, Nozzle Desc: U18"

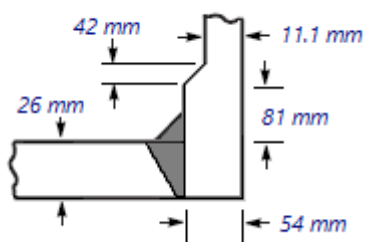
From: 10

Pressure for Reinforcement Calculations	P	28.095	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1407.16	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.16	kgf/cm ²
Inside Diameter of Elliptical Head	D	2025.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	26.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	800.0000	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	81.0000	mm
Height of Beveled Transition	L'	42.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		54.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.



Reinforcement CALCULATION, Description: U18"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 7.750 in.
Actual Thickness Used in Calculation 0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $28.09 \cdot 2028.0 \cdot 0.998 / (2 \cdot 1407.16 \cdot 1.0 - 0.2 \cdot 28.09)$
= 20.2456 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.09 \cdot 99.92 / (1203 \cdot 1.0 - 0.6 \cdot 28.09)$
= 2.3664 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= 42.887/0.5773
= 74.2834 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

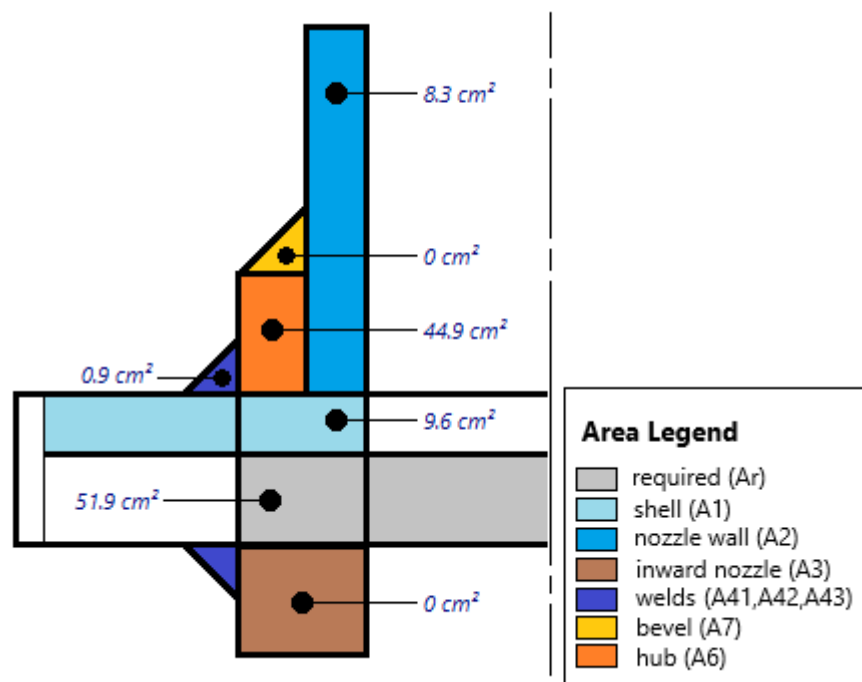
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	482.2970	mm
Parallel to Vessel Wall, opening length	d	241.1485	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	61.2500	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n/S_v)$
= $\min(1, 1203.2/1407.2)$
= 0.855

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n/S_v)$
= $\min(1, 1203.2/1407.2)$
= 0.855

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.855, 1.0)$
= 0.855



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	51.903	NA	NA
Area in Shell	A1	9.612	NA	NA
Area in Nozzle Wall	A2	8.343	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	44.922	NA	NA
TOTAL AREA AVAILABLE	Atot	63.732	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 55.97 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (241.1485 \cdot 20.2456 \cdot 1.0 + 2 \cdot 52.5 \cdot 20.2456 \cdot 1.0 \cdot (1 - 0.86))$
 $= 51.903 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 241.148(1.0 \cdot 24.5 - 1.0 \cdot 20.246) - 2 \cdot 52.5$
 $(1.0 \cdot 24.5 - 1.0 \cdot 20.2456) \cdot (1 - 0.855)$
 $= 9.612 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 61.25)(9.61 - 2.37)0.8551/\sin(65.5)$
 $= 8.343 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$

= 0.855 cm²

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(61.3, 200.0, 81.0)) \cdot (54.0 - 11.1) \cdot 0.855$$

$$= 44.922 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Calculated Minimum Design Metal Temperature -104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -29 °C

Flange MDMT with Temp reduction per UCS-66(i)(2) -49 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 28.09/43.44 = 0.647$$

Weld Size Calculations, Description: U18"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (51.9031 - 9.612 + 2 \cdot 52.5 \cdot 0.8551 \cdot (1.0 \cdot 24.5 - 20.2456)) \cdot 1407)$$

$$= 64885.11 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (53.2651 + 0.8551 - 0.0 \cdot 0.86) \cdot 1407$$

$$= 76155.63 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (53.2651 + 0.0 + 0.8551 + (21.9963)) \cdot 1407$$

$$= 107107.94 \text{ kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (53.2651 + 0.0 + 0.8551 + 0.0 + (21.9963)) \cdot 1407$$

$$= 107107.94 \text{ kgf}$$



Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 367.8465 \cdot 10.0 \cdot 0.49 \cdot 1203$$

$$= 34066. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 152.2487) \cdot (54.0 - 1.5) \cdot 0.7 \cdot 1203$$

$$= 211494. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 367.8465 \cdot (20.0 - 0.0) \cdot 0.74 \cdot 1407$$

$$= 120335. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (34066 + 211494) = 245560 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (34066 + 0 + 120335 + 0) = 154401 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (34066 + 120335 + 0) = 154401 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 245560 kgf, must exceed W = 64885 kgf	or W1 = 76155 kgf
Path 2-2 = 154400 kgf, must exceed W = 64885 kgf	or W2 = 107107 kgf
Path 3-3 = 154400 kgf, must exceed W = 64885 kgf	or W3 = 107107 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 31 kgf/cm²

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: U18"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	7.750 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 28.09 \cdot 2028.0 \cdot 0.998 / (2 \cdot 1407.16 \cdot 1.0 - 0.2 \cdot 28.09)$$

$$= 20.2456 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 28.09 \cdot 99.92 / (1203 \cdot 1.0 - 0.6 \cdot 28.09)$$

$$= 2.3664 \text{ mm}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

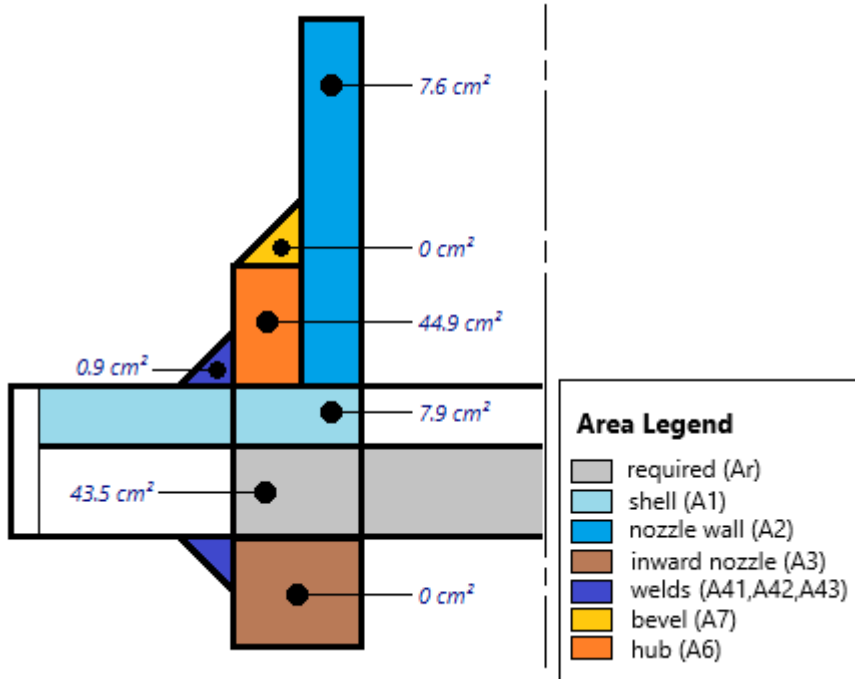
$$= 42.887 / 0.5773$$

$$= 74.2834 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 399.7000 mm
 Parallel to Vessel Wall, opening length d 199.8500 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 61.2500 mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	43.542	NA	NA
Area in Shell	A1	7.855	NA	NA
Area in Nozzle Wall	A2	7.590	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	44.922	NA	NA
TOTAL AREA AVAILABLE	Atot	61.222	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (199.85 \cdot 20.2456 \cdot 1.0 + 2 \cdot 52.5 \cdot 20.2456 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 43.542 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 199.85(1.0 \cdot 24.5 - 1.0 \cdot 20.246) - 2 \cdot 52.5$$

$$(1.0 \cdot 24.5 - 1.0 \cdot 20.2456) \cdot (1 - 0.855)$$

$$= 7.855 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 61.25)(9.61 - 2.37)0.8551$$

= 7.590 cm²

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$$

$$= 0.855 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(61.3, 200.0, 81.0)) \cdot (54.0 - 11.1) \cdot 0.855$$

$$= 44.922 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.8664 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 21.7456 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 21.7456 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 8.6600 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.66, \max(21.7456, 3.0)]$$

$$= 8.6600 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.8664, 8.66)$$

$$= 8.6600 \text{ mm}$$

Available Nozzle Neck Thickness = 11.1125 mm -> OK

Weld Size Calculations, Description: U18"

Intermediate Calc. for nozzle/shell welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (43.542 - 7.855 + 2 \cdot 52.5 \cdot 0.8551 \cdot (1.0 \cdot 24.5 - 20.2456)) \cdot 1407)$$

$$= 55592.09 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - Can/0.707)^2 \cdot fr2) \cdot S_v$$

$$= (52.5122 + 0.8551 - 0.0 \cdot 0.86) \cdot 1407$$

$$= 75096.14 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v$$

$$= (52.5122 + 0.0 + 0.8551 + (21.9963)) \cdot 1407$$

$$= 106048.46 \text{ kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (52.5122 + 0.0 + 0.8551 + 0.0 + (21.9963)) \cdot 1407$$

$$= 106048.46 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\
 &= (3.1416/2.0) \cdot 304.85 \cdot 10.0 \cdot 0.49 \cdot 1203 \\
 &= 28232. \text{ kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\
 &= (3.1416 \cdot 126.175) \cdot (54.0 - 1.5) \cdot 0.7 \cdot 1203 \\
 &= 175274. \text{ kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\
 &= (3.1416/2.0) \cdot 304.85 \cdot (20.0 - 0.0) \cdot 0.74 \cdot 1407 \\
 &= 99726. \text{ kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (28232 + 175274) = 203506 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (28232 + 0 + 99726 + 0) = 127958 \text{ kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (28232 + 99726 + 0) = 127958 \text{ kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 203506 kgf, must exceed W = 55592 kgf	or W1 = 75096 kgf
Path 2-2 = 127958 kgf, must exceed W = 55592 kgf	or W2 = 106048 kgf
Path 3-3 = 127958 kgf, must exceed W = 55592 kgf	or W3 = 106048 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 34 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 82.4724 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 311.2856 mm

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Input, Nozzle Desc: A16"

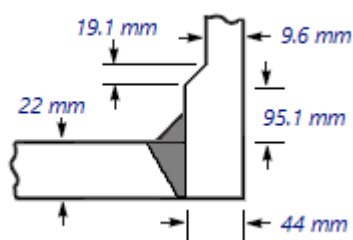
From: 20

Pressure for Reinforcement Calculations	P	28.008	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2025.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		50.00	cm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	95.0683	mm
Height of Beveled Transition	L'	19.0683	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		44.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: A16"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 5.869 in.
 Actual Thickness Used in Calculation 0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.01 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.01)$
 $= 20.4272 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.01 \cdot 76.04 / (1203 \cdot 1.0 - 0.6 \cdot 28.01)$
 $= 1.7951 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 34.399 / 0.5773$
 $= 59.5805 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

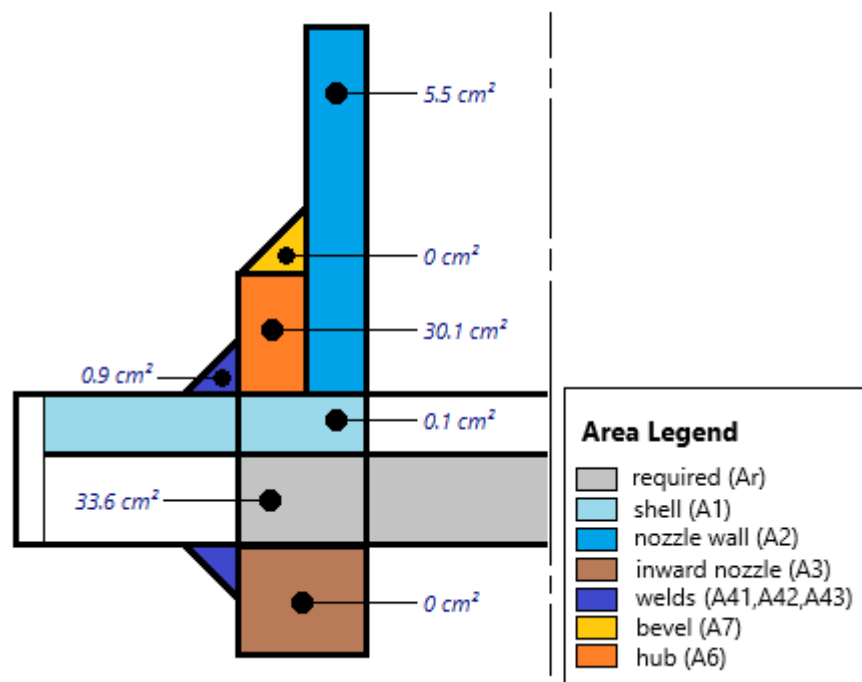
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	304.1452	mm
Parallel to Vessel Wall, opening length	d	152.0726	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203.2 / 1407.1)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203.2 / 1407.1)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	33.581	NA	NA
Area in Shell	A1	0.102	NA	NA
Area in Nozzle Wall	A2	5.527	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	30.149	NA	NA
TOTAL AREA AVAILABLE	Atot	36.633	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (152.0726 \cdot 20.4272 \cdot 1.0 + 2 \cdot 42.5 \cdot 20.4272 \cdot 1.0 \cdot (1 - 0.86))$
 $= 33.581 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 152.073(1.0 \cdot 20.5 - 1.0 \cdot 20.427) - 2 \cdot 42.5$
 $(1.0 \cdot 20.5 - 1.0 \cdot 20.4272) \cdot (1 - 0.855)$
 $= 0.102 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 51.25)(8.1 - 1.8)0.8551$
 $= 5.527 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$
 $= 0.855 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(51.2, 400.0, 95.1)) \cdot (44.0 - 9.6) \cdot 0.855$$

$$= 30.149 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 3.2951 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 21.9272 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 21.9272 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 7.7200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.72, \max(21.9272, 3.0)]$
 $= 7.7200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.2951, 7.72)$
 $= 7.7200 \text{ mm}$

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	410.0,	Allowable	:	1203.2 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2598.0 kgf/cm ²	Passed
Occasional	:	124.8,	Allowable	:	1600.3 kgf/cm ²	Passed
Shear	:	212.9,	Allowable	:	842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 28.01/86.88 = 0.322

Weld Size Calculations, Description: A16"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (33.5805 - 0.1018 + 2 \cdot 42.5 \cdot 0.8551 \cdot (1.0 \cdot 20.5 - 20.4272)) \cdot 1407)$
 $= 47183.54 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (35.6758 + 0.8551 - 0.0 \cdot 0.86) \cdot 1407$
 $= 51403.79 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (35.6758 + 0.0 + 0.8551 + (14.8996)) \cdot 1407$
 $= 72369.55 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (35.6758 + 0.0 + 0.8551 + 0.0 + (14.8996)) \cdot 1407$
 $= 72369.55 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 237.0726 \cdot 10.0 \cdot 0.49 \cdot 1203$
 $= 21955. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 97.2863) \cdot (44.0 - 1.5) \cdot 0.7 \cdot 1203$
 $= 109402. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 237.0726 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 1407$
 $= 79492. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (21955 + 109402) = 131357 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (21955 + 0 + 79492 + 0) = 101447 \text{ kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (21955 + 79492 + 0) = 101447 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 131357 kgf, must exceed W = 47183 kgf	or W1 = 51403 kgf
Path 2-2 = 101446 kgf, must exceed W = 47183 kgf	or W2 = 72369 kgf
Path 3-3 = 101446 kgf, must exceed W = 47183 kgf	or W3 = 72369 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 28 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 6.9626 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 428.9626 mm

Input Echo, WRC297 Item 1, Description: A16" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2025.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	149.073	mm
Nozzle Thickness	44.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF1	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	28.01	kgf/cm ²
Radial Load	P	-978.93	kgf
Circumferential Shear	Vc	734.20	kgf
Longitudinal Shear	Vl	978.93	kgf
Circumferential Moment	Mc	293.68	kgf-m.
Longitudinal Moment	Ml	381.78	kgf-m.
Torsional Moment	Mt	440.52	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (5.388) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 237.073 + 2 \cdot 1.65 \cdot \sqrt{(1024.25(22.0 - 1.5))}$
 $= 715.255 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2048.500 mm
Nozzle Outside Diameter	(d) =	237.073 mm
Vessel Thickness used	(T) =	20.500 mm
Nozzle Thickness used	(t) =	42.500 mm
T / t	=	0.482
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.157$$

Nr/P	=	0.097
Mr/P	=	0.204



M0/P = 0.060
 N0/P = 0.125
 MrD/Mc = 0.305
 NrDL/Mc = 0.092
 M0d/Mc = 0.095
 N0DL/Mc = 0.046
 MrD/Ml = 0.177
 NrDL/Ml = 0.085
 M0D/Ml = 0.057
 N0DL/Ml = 0.126

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	29	29	29	29
Outplane Bending (P)	83	-83	83	-83
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-48	-48	48	48
Outplane Bending (ML)	-131	131	131	-131
Normal Pressure Stress	1371	1399	1371	1399
Outplane Stress Summary	1304	1427	1663	1262

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	22	22	22	22
Inplane Bending (P)	285	-285	285	-285
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-32	-32	32	32
Inplane Bending (ML)	-407	407	407	-407
Inplane Pressure Stress	685	685	685	685
Inplane Stress Summary	554	797	1433	48

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	9	9	-9	-9
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	24	24	24	24
Shear Stress Summary	33	33	14	14

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	1305	1429	1664	1262

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	29	29	29	29
Outplane Bending (P)	83	-83	83	-83
Outplane Membrane (Mc)	-13	-13	13	13
Outplane Bending (Mc)	-167	167	167	-167
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0

Normal Pressure Stress	685	685	685	685
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Outplane Stress Summary	616	785	979	477
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	22	22	22	22
Inplane Bending (P)	285	-285	285	-285
Inplane Membrane (Mc)	-27	-27	27	27
Inplane Bending (Mc)	-539	539	539	-539
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1371	1399	1371	1399

Inplane Stress Summary	1113	1648	2245	624
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	24	24	24	24

Shear Stress Summary	11	11	37	37
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	1113	1648	2246	633
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	29	29	29	29
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-48	-48	48	48
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	46	80	46	80

Hoop Stress Summary	27	61	124	157
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	3	3	3	3
Axial Bending (P)	50	-50	50	-50
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-35	-35	35	35
Axial Bending (ML)	-71	71	71	-71
Axial Pressure Stress	23	23	23	23

Axial Stress Summary	-29	13	184	-60
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11

Shear Stress Summary	16	16	7	7
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	65	66	185	218
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	29	29	29	29
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-13	-13	13	13
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	46	80	46	80

Hoop Stress Summary	62	95	89	123
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	3	3	3	3
Axial Bending (P)	50	-50	50	-50
Axial Membrane (Mc)	-26	-26	26	26
Axial Bending (Mc)	-106	106	106	-106
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	23	23	23	23

Axial Stress Summary	-55	55	211	-102
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-6	-6	6	6
Shear due to Torsion	11	11	11	11

Shear Stress Summary	5	5	17	17
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	118	96	213	228
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1371	1399	1371	1399	1371	1399	1371	1399
Circ. Pl (SUS)	-19	-19	77	77	-5	-5	49	49
Circ. Q (SUS)	-48	48	214	-214	-254	254	824	-824

Long. Pm (SUS)	685	685	685	685	685	685	685	685
Long. Pl (SUS)	-10	-10	54	54	16	16	42	42
Long. Q (SUS)	-122	122	692	-692	-84	84	250	-250
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	9	9	-9	-9	-12	-12	12	12
Shear Q (SUS)	24	24	24	24	24	24	24	24
Pm (SUS)	1371	1399	1371	1399	1371	1399	1371	1399
Pm+Pl (SUS)	1352	1380	1448	1476	1366	1394	1420	1448
Pm+Pl+Q (Total)	1305	1429	1662	1262	1112	1648	2245	632

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1399.00	1407.13	Passed
Pm+Pl (SUS)	1476.11	2110.70	Passed
Pm+Pl+Q (TOTAL)	2245.02	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	46	80	46	80	46	80	46	80
Circ. Pl (SUS)	-19	-19	77	77	16	16	42	42
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. Pl (SUS)	-32	-32	38	38	-23	-23	29	29
Long. Q (SUS)	-21	21	121	-121	-56	56	156	-156
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-6	-6	6	6
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	46.0	80.0	46.0	80.0	46.0	80.0	46.0	80.0
Pm+Pl (SUS)	36.9	70.5	123.3	157.2	63.2	96.7	89.0	122.5
Pm+Pl+Q (Total)	64.4	65.2	182.8	217.5	118.4	96.6	210.4	228.5

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	80.00	1203.20	Passed
Pm+Pl (SUS)	157.17	1804.80	Passed
Pm+Pl+Q (TOTAL)	228.54	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Input, Nozzle Desc: A26"

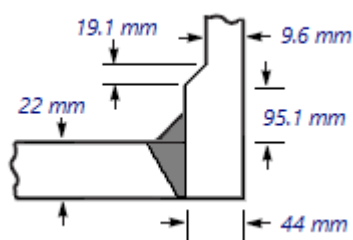
From: 20

Pressure for Reinforcement Calculations	P	28.008	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2025.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		140.00	cm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	95.0683	mm
Height of Beveled Transition	L'	19.0683	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		44.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: A26"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 5.869 in.
Actual Thickness Used in Calculation 0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.01 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.01)$
= 20.4272 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.01 \cdot 76.04 / (1203 \cdot 1.0 - 0.6 \cdot 28.01)$
= 1.7951 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $34.399 / 0.5773$
= 59.5805 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

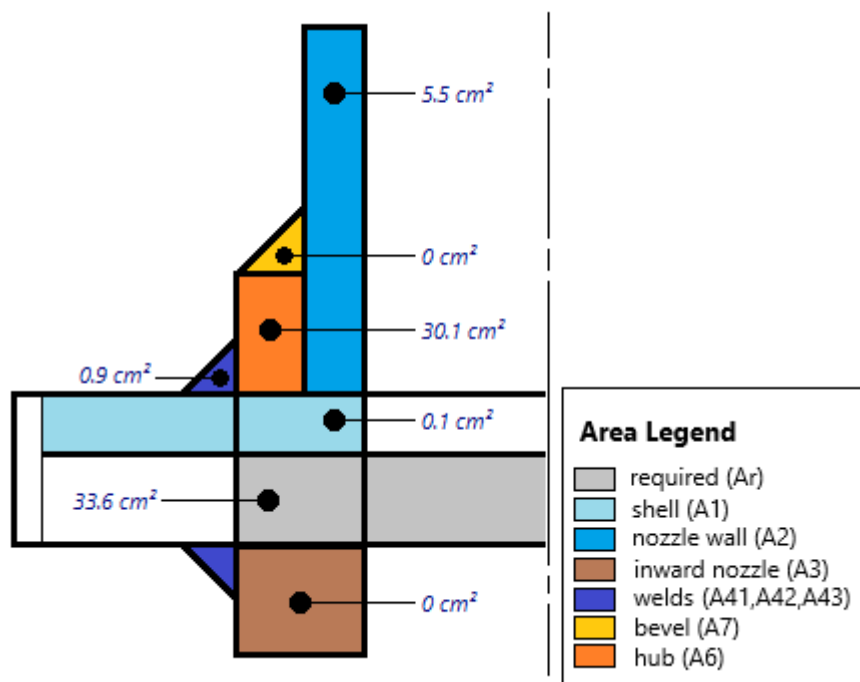
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	304.1452	mm
Parallel to Vessel Wall, opening length	d	152.0726	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 1203.2 / 1407.1)$
= 0.855

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 1203.2 / 1407.1)$
= 0.855

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.855, 1.0)$
= 0.855



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	33.581	NA	NA
Area in Shell	A1	0.102	NA	NA
Area in Nozzle Wall	A2	5.527	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	30.149	NA	NA
TOTAL AREA AVAILABLE	Atot	36.633	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (152.0726 \cdot 20.4272 \cdot 1.0 + 2 \cdot 42.5 \cdot 20.4272 \cdot 1.0 \cdot (1 - 0.86))$
 $= 33.581 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 152.073(1.0 \cdot 20.5 - 1.0 \cdot 20.427) - 2 \cdot 42.5$
 $(1.0 \cdot 20.5 - 1.0 \cdot 20.4272) \cdot (1 - 0.855)$
 $= 0.102 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 51.25)(8.1 - 1.8)0.8551$
 $= 5.527 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$
 $= 0.855 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(51.2, 400.0, 95.1)) \cdot (44.0 - 9.6) \cdot 0.855$$

$$= 30.149 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 3.2951 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 21.9272 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 21.9272 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 7.7200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.72, \max(21.9272, 3.0)]$$

$$= 7.7200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.2951, 7.72)$$

$$= 7.7200 \text{ mm}$$

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	410.0,	Allowable	:	1203.2 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2598.0 kgf/cm ²	Passed
Occasional	:	124.8,	Allowable	:	1600.3 kgf/cm ²	Passed
Shear	:	212.9,	Allowable	:	842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $28.01/86.88 = 0.322$

Weld Size Calculations, Description: A26"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (33.5805 - 0.1018 + 2 \cdot 42.5 \cdot 0.8551 \cdot (1.0 \cdot 20.5 - 20.4272)) \cdot 1407)$
 $= 47183.54 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (35.6758 + 0.8551 - 0.0 \cdot 0.86) \cdot 1407$
 $= 51403.79 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (35.6758 + 0.0 + 0.8551 + (14.8996)) \cdot 1407$
 $= 72369.55 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (35.6758 + 0.0 + 0.8551 + 0.0 + (14.8996)) \cdot 1407$
 $= 72369.55 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 237.0726 \cdot 10.0 \cdot 0.49 \cdot 1203$
 $= 21955. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 97.2863) \cdot (44.0 - 1.5) \cdot 0.7 \cdot 1203$
 $= 109402. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 237.0726 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 1407$
 $= 79492. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (21955 + 109402) = 131357 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (21955 + 0 + 79492 + 0) = 101447 \text{ kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (21955 + 79492 + 0) = 101447 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 131357 kgf, must exceed W = 47183 kgf	or W1 = 51403 kgf
Path 2-2 = 101446 kgf, must exceed W = 47183 kgf	or W2 = 72369 kgf
Path 3-3 = 101446 kgf, must exceed W = 47183 kgf	or W3 = 72369 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 28 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 6.9626 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 428.9626 mm

Input Echo, WRC297 Item 1, Description: A26" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2025.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	149.073	mm
Nozzle Thickness	44.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF1	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	28.01	kgf/cm ²
Radial Load	P	-978.93	kgf
Circumferential Shear	Vc	734.20	kgf
Longitudinal Shear	Vl	978.93	kgf
Circumferential Moment	Mc	293.68	kgf-m.
Longitudinal Moment	Ml	381.78	kgf-m.
Torsional Moment	Mt	440.52	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (5.388) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 237.073 + 2 \cdot 1.65 \cdot \sqrt{(1024.25(22.0 - 1.5))}$
 $= 715.255 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2048.500 mm
Nozzle Outside Diameter	(d) =	237.073 mm
Vessel Thickness used	(T) =	20.500 mm
Nozzle Thickness used	(t) =	42.500 mm
T / t	=	0.482
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.157$$

Nr/P	=	0.097
Mr/P	=	0.204



M0/P = 0.060
 N0/P = 0.125
 MrD/Mc = 0.305
 NrDL/Mc = 0.092
 M0d/Mc = 0.095
 N0DL/Mc = 0.046
 MrD/Ml = 0.177
 NrDL/Ml = 0.085
 M0D/Ml = 0.057
 N0DL/Ml = 0.126

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	29	29	29	29
Outplane Bending (P)	83	-83	83	-83
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-48	-48	48	48
Outplane Bending (ML)	-131	131	131	-131
Normal Pressure Stress	1371	1399	1371	1399
Outplane Stress Summary	1304	1427	1663	1262

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	22	22	22	22
Inplane Bending (P)	285	-285	285	-285
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-32	-32	32	32
Inplane Bending (ML)	-407	407	407	-407
Inplane Pressure Stress	685	685	685	685
Inplane Stress Summary	554	797	1433	48

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	9	9	-9	-9
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	24	24	24	24
Shear Stress Summary	33	33	14	14

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1305	1429	1664	1262

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	29	29	29	29
Outplane Bending (P)	83	-83	83	-83
Outplane Membrane (Mc)	-13	-13	13	13
Outplane Bending (Mc)	-167	167	167	-167
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



Normal Pressure Stress	685	685	685	685
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Outplane Stress Summary	616	785	979	477
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	22	22	22	22
Inplane Bending (P)	285	-285	285	-285
Inplane Membrane (Mc)	-27	-27	27	27
Inplane Bending (Mc)	-539	539	539	-539
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1371	1399	1371	1399

Inplane Stress Summary	1113	1648	2245	624
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	24	24	24	24

Shear Stress Summary	11	11	37	37
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	1113	1648	2246	633
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	29	29	29	29
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-48	-48	48	48
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	46	80	46	80

Hoop Stress Summary	27	61	124	157
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	3	3	3	3
Axial Bending (P)	50	-50	50	-50
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-35	-35	35	35
Axial Bending (ML)	-71	71	71	-71
Axial Pressure Stress	23	23	23	23

Axial Stress Summary	-29	13	184	-60
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11

Shear Stress Summary	16	16	7	7
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	65	66	185	218
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	29	29	29	29
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-13	-13	13	13
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	46	80	46	80

Hoop Stress Summary	62	95	89	123
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	3	3	3	3
Axial Bending (P)	50	-50	50	-50
Axial Membrane (Mc)	-26	-26	26	26
Axial Bending (Mc)	-106	106	106	-106
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	23	23	23	23

Axial Stress Summary	-55	55	211	-102
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-6	-6	6	6
Shear due to Torsion	11	11	11	11

Shear Stress Summary	5	5	17	17
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	118	96	213	228
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1371	1399	1371	1399	1371	1399	1371	1399
Circ. Pl (SUS)	-19	-19	77	77	-5	-5	49	49
Circ. Q (SUS)	-48	48	214	-214	-254	254	824	-824



Long. Pm (SUS)	685	685	685	685	685	685	685	685
Long. Pl (SUS)	-10	-10	54	54	16	16	42	42
Long. Q (SUS)	-122	122	692	-692	-84	84	250	-250
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	9	9	-9	-9	-12	-12	12	12
Shear Q (SUS)	24	24	24	24	24	24	24	24
Pm (SUS)	1371	1399	1371	1399	1371	1399	1371	1399
Pm+Pl (SUS)	1352	1380	1448	1476	1366	1394	1420	1448
Pm+Pl+Q (Total)	1305	1429	1662	1262	1112	1648	2245	632

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1399.00	1407.13	Passed
Pm+Pl (SUS)	1476.11	2110.70	Passed
Pm+Pl+Q (TOTAL)	2245.02	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	46	80	46	80	46	80	46	80
Circ. Pl (SUS)	-19	-19	77	77	16	16	42	42
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. Pl (SUS)	-32	-32	38	38	-23	-23	29	29
Long. Q (SUS)	-21	21	121	-121	-56	56	156	-156
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-6	-6	6	6
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	46.0	80.0	46.0	80.0	46.0	80.0	46.0	80.0
Pm+Pl (SUS)	36.9	70.5	123.3	157.2	63.2	96.7	89.0	122.5
Pm+Pl+Q (Total)	64.4	65.2	182.8	217.5	118.4	96.6	210.4	228.5

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	80.00	1203.20	Passed
Pm+Pl (SUS)	157.17	1804.80	Passed
Pm+Pl+Q (TOTAL)	228.54	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: P12"

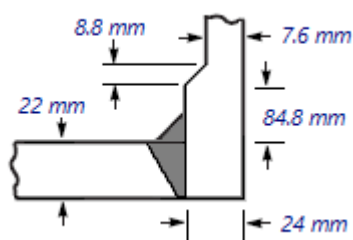
From: 20

Pressure for Reinforcement Calculations	P	28.008	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2025.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		205.00	cm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	84.8118	mm
Height of Beveled Transition	L'	8.8118	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: P12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.01 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.01)$
 $= 20.4272$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.01 \cdot 24.02 / (1203 \cdot 1.0 - 0.6 \cdot 28.01)$
 $= 0.5670$ mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 22.5$, less than the hub height, use (e-2)
 $= 56.2500$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	134.0342	mm
Parallel to Vessel Wall	Rn+tn+t	67.0171	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: P12".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0670	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 21.9272	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 21.9272	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(21.9272, 3.0)]$
 $= 4.9200$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.067, 4.92)$
 $= 4.9200$ mm

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Weld Size Calculations, Description: P12"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0691 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 423.0691 mm

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Input, Nozzle Desc: P22"

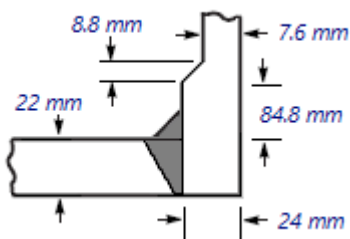
From: 20

Pressure for Reinforcement Calculations	P	28.008	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2025.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		255.00	cm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	84.8118	mm
Height of Beveled Transition	L'	8.8118	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: P22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.01 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.01)$
 $= 20.4272$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.01 \cdot 24.02 / (1203 \cdot 1.0 - 0.6 \cdot 28.01)$
 $= 0.5670$ mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 22.5$, less than the hub height, use (e-2)
 $= 56.2500$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	134.0342	mm
Parallel to Vessel Wall	Rn+tn+t	67.0171	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: P22".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0670	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 21.9272	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 21.9272	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(21.9272, 3.0)]$
 $= 4.9200$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.067, 4.92)$
 $= 4.9200$ mm

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Weld Size Calculations, Description: P22"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0691 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 423.0691 mm

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Input, Nozzle Desc: A32"

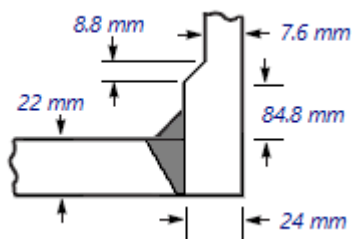
From: 20

Pressure for Reinforcement Calculations	P	28.107	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2025.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		205.00	cm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	84.8118	mm
Height of Beveled Transition	L'	8.8118	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: A32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.11 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.11)$
 $= 20.4999$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.11 \cdot 24.02 / (1203 \cdot 1.0 - 0.6 \cdot 28.11)$
 $= 0.5690$ mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot$ Corroded Hub Thickness
 $= 2.5 \cdot 22.5$, less than the hub height, use (e-2)
 $= 56.2500$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	134.0342	mm
Parallel to Vessel Wall	Rn+tn+t	67.0171	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: A32".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0690	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 21.9999	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 21.9999	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(22.0, 3.0)]$
 $= 4.9200$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.069, 4.92)$
 $= 4.9200$ mm

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 432.7,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2575.3 kgf/cm ²	Passed
Occasional	: 48.7,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 307.7,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :



Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: A32"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0691 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.0691 mm

Input Echo, WRC297 Item 1, Description: A32" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2025.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1407.13 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13 kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	45.034	mm
Nozzle Thickness	24.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF1	
Nozzle Cold S.I. Allowable	SNmc	1203.20 kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1203.20 kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	28.11	kgf/cm ²
Radial Load	P	-326.31	kgf
Circumferential Shear	Vc	244.73	kgf
Longitudinal Shear	Vl	326.31	kgf
Circumferential Moment	Mc	32.63	kgf-m.
Longitudinal Moment	Ml	42.42	kgf-m.
Torsional Moment	Mt	48.95	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (4.229) is < 5
 check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (3.876) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 93.034 + 2 \cdot 1.65 \cdot \sqrt{(1024.25(22.0 - 1.5))}$$

$$= 571.217 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2048.500 mm
 Nozzle Outside Diameter (d) = 93.034 mm
 Vessel Thickness used (T) = 20.500 mm
 Nozzle Thickness used (t) = 22.500 mm
 T / t = 0.911
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.454$$

Nr/P = 0.176
 Mr/P = 0.302
 M0/P = 0.103
 N0/P = 0.180
 MrD/Mc = 0.295
 NrDL/Mc = 0.131
 M0d/Mc = 0.115
 N0DL/Mc = -0.093
 MrD/Ml = 0.268
 NrDL/Ml = 0.130
 M0D/Ml = 0.102
 N0DL/Ml = -0.010

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	47	-47	47	-47
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	1	1	-1	-1
Outplane Bending (ML)	-66	66	66	-66
Normal Pressure Stress	1376	1404	1376	1404
Outplane Stress Summary	1372	1438	1503	1302

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	13	13	13	13
Inplane Bending (P)	140	-140	140	-140
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-14	-14	14	14
Inplane Bending (ML)	-174	174	174	-174
Inplane Pressure Stress	688	688	688	688
Inplane Stress Summary	653	721	1031	400

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	8	8	-8	-8
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	17	17	17	17
Shear Stress Summary	25	25	9	9

**Vessel Stresses (kgf/cm²):**

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1373	1438	1503	1302

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	47	-47	47	-47
Outplane Membrane (Mc)	7	7	-7	-7
Outplane Bending (Mc)	-57	57	57	-57
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	688	688	688	688
Outplane Stress Summary	700	719	799	588

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	13	13	13	13
Inplane Bending (P)	140	-140	140	-140
Inplane Membrane (Mc)	-10	-10	10	10
Inplane Bending (Mc)	-147	147	147	-147
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1376	1404	1376	1404
Inplane Stress Summary	1371	1414	1689	1140

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-10	-10	10	10
Torsional Shear (Mt)	17	17	17	17
Shear Stress Summary	6	6	28	28

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1371	1414	1689	1142

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	13	13	13	13
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	1	1	-1	-1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	24	58	24	58
Hoop Stress Summary	39	73	37	71

**Nozzle Stresses (kgf/cm²):**

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	6	6	6	6
Axial Bending (P)	82	-82	82	-82
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-57	-57	57	57
Axial Bending (ML)	-109	109	109	-109
Axial Pressure Stress	12	12	12	12
Axial Stress Summary	-66	-11	268	-115

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	B1
	Outside	Inside	Outside	Inside
Shear due to (Vc)	7	7	-7	-7
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	15	15	15	15
Shear Stress Summary	23	23	8	8

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	115	96	269	187

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	D1
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	13	13	13	13
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	7	7	-7	-7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	24	58	24	58
Hoop Stress Summary	46	79	30	64

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	D1
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	6	6	6	6
Axial Bending (P)	82	-82	82	-82
Axial Membrane (Mc)	-44	-44	44	44
Axial Bending (Mc)	-95	95	95	-95
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	12	12	12	12
Axial Stress Summary	-38	-13	240	-114

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	D1
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-9	-9	9	9
Shear due to Torsion	15	15	15	15

Shear Stress Summary 6 6 25 25

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	85	93	244	186

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1376	1404	1376	1404	1376	1404	1376	1404
Circ. Pl (SUS)	14	14	12	12	3	3	23	23
Circ. Q (SUS)	-19	19	113	-113	-7	7	287	-287
Long. Pm (SUS)	688	688	688	688	688	688	688	688
Long. Pl (SUS)	-1	-1	27	27	20	20	6	6
Long. Q (SUS)	-34	34	314	-314	-10	10	104	-104
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	8	8	-8	-8	-10	-10	10	10
Shear Q (SUS)	17	17	17	17	17	17	17	17
Pm (SUS)	1376	1404	1376	1404	1376	1404	1376	1404
Pm+Pl (SUS)	1390	1418	1388	1416	1379	1407	1399	1427
Pm+Pl+Q (Total)	1371	1437	1501	1303	1372	1414	1686	1141

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1404.00	1407.13	Passed
Pm+Pl (SUS)	1427.14	2110.70	Passed
Pm+Pl+Q (TOTAL)	1686.82	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	24	58	24	58	24	58	24	58
Circ. Pl (SUS)	14	14	12	12	20	20	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. Pl (SUS)	-51	-51	63	63	-38	-38	50	50
Long. Q (SUS)	-27	27	191	-191	-13	13	177	-177
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-9	-9	9	9
Shear Q (SUS)	15	15	15	15	15	15	15	15
Pm (SUS)	24.0	58.0	24.0	58.0	24.0	58.0	24.0	58.0
Pm+Pl (SUS)	78.3	111.9	76.2	79.9	72.3	105.5	64.4	72.1
Pm+Pl+Q (Total)	112.9	94.8	266.3	186.7	83.9	91.8	241.7	185.3

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	58.00	1203.20	Passed
Pm+Pl (SUS)	111.88	1804.80	Passed
Pm+Pl+Q (TOTAL)	266.28	3609.61	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: L32"

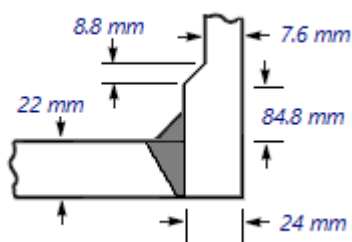
From: 20

Pressure for Reinforcement Calculations	P	28.020	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2025.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	500.0000	mm
Distance from Bottom/Left Tangent		155.00	cm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		28.90	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	84.8118	mm
Height of Beveled Transition	L'	8.8118	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: L32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.02 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.02)$
 $= 20.4357 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.02 \cdot 24.02 / (1203 \cdot 1.0 - 0.6 \cdot 28.02)$
 $= 0.5672 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 22.5$, less than the hub height, use (e-2)
 $= 56.2500 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	141.0510	mm
Parallel to Vessel Wall	Rn+tn+t	70.5255	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L32".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: L32"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: L32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.02 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.02)$
 $= 20.4357$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.02 \cdot 24.02 / (1203 \cdot 1.0 - 0.6 \cdot 28.02)$
 $= 0.5672$ mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 22.5$, less than the hub height, use (e-2)
 $= 56.2500$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	134.0342	mm
Parallel to Vessel Wall	Rn+tn+tt	67.0171	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L32".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0672	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 21.9357	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 21.9357	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(21.9357, 3.0)]$
 $= 4.9200$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.0672, 4.92)$
 $= 4.9200$ mm

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Weld Size Calculations, Description: L32"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 28.0943 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 253.3084 mm

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Input, Nozzle Desc: L42"

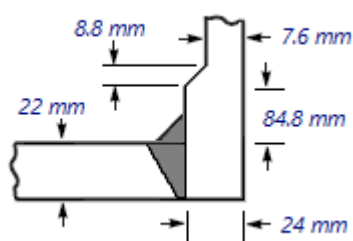
From: 20

Pressure for Reinforcement Calculations	P	28.077	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2025.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	500.0000	mm
Distance from Bottom/Left Tangent		155.00	cm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-28.90	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	84.8118	mm
Height of Beveled Transition	L'	8.8118	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: L42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.773 in.
Actual Thickness Used in Calculation	0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.08 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.08)$
 $= 20.4775 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.08 \cdot 24.02 / (1203 \cdot 1.0 - 0.6 \cdot 28.08)$
 $= 0.5684 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 22.5$, less than the hub height, use (e-2)
 $= 56.2500 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	141.0506 mm
Parallel to Vessel Wall	Rn+tn+t	70.5253 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L42".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Weld Size Calculations, Description: L42"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: L42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.773 in.
Actual Thickness Used in Calculation	0.301 in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.08 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.08)$
 $= 20.4775 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.08 \cdot 24.02 / (1203 \cdot 1.0 - 0.6 \cdot 28.08)$
 $= 0.5684 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 22.5$, less than the hub height, use (e-2)
 $= 56.2500 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	134.0342	mm
Parallel to Vessel Wall	Rn+tn+tt	67.0171	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L42".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0684	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 21.9775	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 21.9775	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(21.9775, 3.0)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.0684, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Weld Size Calculations, Description: L42"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 28.0943 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 253.3084 mm

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Input, Nozzle Desc: S13"

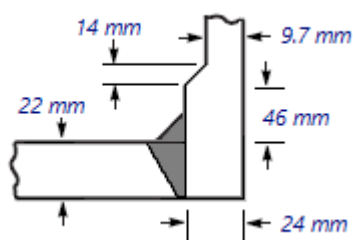
From: 30

Pressure for Reinforcement Calculations	P	28.008	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2025.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		455.00	cm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	46.0000	mm
Height of Beveled Transition	L'	14.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: S13"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.734 in.
Actual Thickness Used in Calculation 0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.01 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.01)$
= 20.4272 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.01 \cdot 36.22 / (1203 \cdot 1.0 - 0.6 \cdot 28.01)$
= 0.8550 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 14.265/0.5773
= 24.7085 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

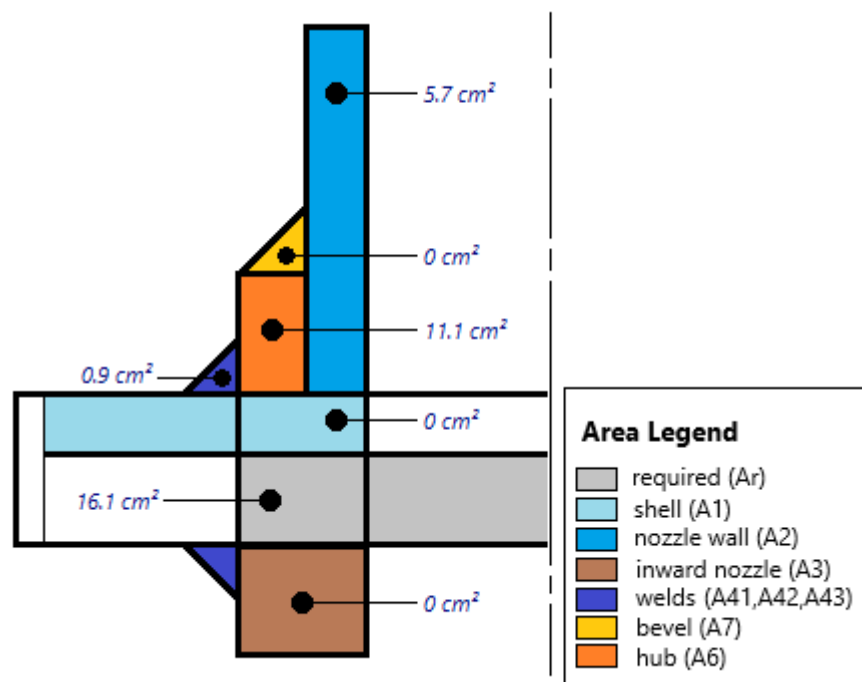
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	144.8618	mm
Parallel to Vessel Wall, opening length	d	72.4309	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	45.2949	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 1203.2/1407.1)
= 0.855

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 1203.2/1407.1)
= 0.855

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.855, 1.0)
= 0.855



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	16.128	NA	NA
Area in Shell	A1	0.048	NA	NA
Area in Nozzle Wall	A2	5.716	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	11.050	NA	NA
TOTAL AREA AVAILABLE	Atot	17.669	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (72.4309 \cdot 20.4272 \cdot 1.0 + 2 \cdot 22.5 \cdot 20.4272 \cdot 1.0 \cdot (1 - 0.86))$
 $= 16.128 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 72.431(1.0 \cdot 20.5 - 1.0 \cdot 20.427) - 2 \cdot 22.5$
 $(1.0 \cdot 20.5 - 1.0 \cdot 20.4272) \cdot (1 - 0.855)$
 $= 0.048 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 45.29)(8.23 - 0.85) \cdot 0.8551$
 $= 5.716 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$
 $= 0.855 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(45.3, 400.0, 46.0)) \cdot (24.0 - 9.7) \cdot 0.855$$

$$= 11.050 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.3550 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 21.9272 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 21.9272 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 6.3000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.3, \max(21.9272, 3.0)]$$

$$= 6.3000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.355, 6.3)$$

$$= 6.3000 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	343.9,	Allowable	:	1203.2 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2664.1 kgf/cm ²	Passed
Occasional	:	55.3,	Allowable	:	1600.3 kgf/cm ²	Passed
Shear	:	227.6,	Allowable	:	842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 28.01/86.88 = 0.322

Weld Size Calculations, Description: S13"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (16.1278 - 0.048 + 2 \cdot 22.5 \cdot 0.8551 \cdot (1.0 \cdot 20.5 - 20.4272)) \cdot 1407)$
 $= 22665.84 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (16.7664 + 0.8551 - 0.0 \cdot 0.86) \cdot 1407$
 $= 24795.76 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (16.7664 + 0.0 + 0.8551 + (7.888)) \cdot 1407$
 $= 35895.28 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (16.7664 + 0.0 + 0.8551 + 0.0 + (7.888)) \cdot 1407$
 $= 35895.28 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 117.4309 \cdot 10.0 \cdot 0.49 \cdot 1203$
 $= 10875. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 47.4655) \cdot (24.0 - 1.5) \cdot 0.7 \cdot 1203$
 $= 28258. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 117.4309 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 1407$
 $= 39375. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (10875 + 28258) = 39133 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (10875 + 0 + 39375 + 0) = 50250 \text{ kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (10875 + 39375 + 0) = 50250 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 39133	kgf, must exceed W = 22665 kgf	or W1 = 24795 kgf
Path 2-2 = 50250	kgf, must exceed W = 22665 kgf	or W2 = 35895 kgf
Path 3-3 = 50250	kgf, must exceed W = 22665 kgf	or W3 = 35895 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 28 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.7039 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 423.7039 mm

Input Echo, WRC297 Item 1, Description: S13" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2025.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	69.431	mm
Nozzle Thickness	24.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF1	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	28.01	kgf/cm ²
Radial Load	P	-489.47	kgf
Circumferential Shear	Vc	367.10	kgf
Longitudinal Shear	Vl	489.47	kgf
Circumferential Moment	Mc	73.42	kgf-m.
Longitudinal Moment	Ml	95.45	kgf-m.
Torsional Moment	Mt	110.13	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (4.893) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 117.431 + 2 \cdot 1.65 \cdot \sqrt{(1024.25(22.0 - 1.5))}$
 $= 595.614 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2048.500 mm
Nozzle Outside Diameter	(d) =	117.431 mm
Vessel Thickness used	(T) =	20.500 mm
Nozzle Thickness used	(t) =	22.500 mm
T / t	=	0.911
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.573$$

Nr/P	=	0.161
Mr/P	=	0.268



M0/P = 0.091
 N0/P = 0.164
 MrD/Mc = 0.293
 NrDL/Mc = 0.133
 M0d/Mc = 0.113
 N0DL/Mc = -0.081
 MrD/M1 = 0.252
 NrDL/M1 = 0.128
 M0D/M1 = 0.096
 N0DL/M1 = 0.020

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	18	18	18	18
Outplane Bending (P)	63	-63	63	-63
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-111	111	111	-111
Normal Pressure Stress	1371	1399	1371	1399
Outplane Stress Summary	1338	1462	1568	1247

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	18	18	18	18
Inplane Bending (P)	187	-187	187	-187
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-24	-24	24	24
Inplane Bending (ML)	-292	292	292	-292
Inplane Pressure Stress	685	685	685	685
Inplane Stress Summary	574	784	1208	249

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	9	9	-9	-9
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	24	24	24	24
Shear Stress Summary	34	34	15	15

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1340	1464	1569	1247

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	18	18	18	18
Outplane Bending (P)	63	-63	63	-63
Outplane Membrane (Mc)	11	11	-11	-11
Outplane Bending (Mc)	-101	101	101	-101
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



Normal Pressure Stress	685	685	685	685
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Outplane Stress Summary	678	754	857	528
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	18	18	18	18
Inplane Bending (P)	187	-187	187	-187
Inplane Membrane (Mc)	-19	-19	19	19
Inplane Bending (Mc)	-261	261	261	-261
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1371	1399	1371	1399

Inplane Stress Summary	1296	1472	1859	988
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	24	24	24	24

Shear Stress Summary	11	11	37	37
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	1296	1472	1860	991
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	18	18	18	18
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	41	74	41	74

Hoop Stress Summary	56	90	64	97
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	7	7	7	7
Axial Bending (P)	108	-108	108	-108
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-70	-70	70	70
Axial Bending (ML)	-181	181	181	-181
Axial Pressure Stress	20	20	20	20

Axial Stress Summary	-114	29	388	-192
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	8	8	-8	-8
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	22	22	22	22

Shear Stress Summary	31	31	13	13
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	182	103	388	291
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	18	18	18	18
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	11	11	-11	-11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	41	74	41	74

Hoop Stress Summary	72	105	48	81
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	7	7	7	7
Axial Bending (P)	108	-108	108	-108
Axial Membrane (Mc)	-53	-53	53	53
Axial Bending (Mc)	-167	167	167	-167
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	20	20	20	20

Axial Stress Summary	-85	32	358	-194
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-11	-11	11	11
Shear due to Torsion	22	22	22	22

Shear Stress Summary	10	10	34	34
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	158	107	362	285
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1371	1399	1371	1399	1371	1399	1371	1399
Circ. Pl (SUS)	15	15	21	21	-1	-1	37	37
Circ. Q (SUS)	-48	48	174	-174	-74	74	448	-448

Long. Pm (SUS)	685	685	685	685	685	685	685	685
Long. Pl (SUS)	-6	-6	42	42	29	29	7	7
Long. Q (SUS)	-105	105	479	-479	-38	38	164	-164
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	9	9	-9	-9	-12	-12	12	12
Shear Q (SUS)	24	24	24	24	24	24	24	24
Pm (SUS)	1371	1399	1371	1399	1371	1399	1371	1399
Pm+Pl (SUS)	1386	1414	1392	1420	1370	1398	1408	1436
Pm+Pl+Q (Total)	1339	1463	1566	1246	1296	1472	1857	990

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1399.00	1407.13	Passed
Pm+Pl (SUS)	1436.19	2110.70	Passed
Pm+Pl+Q (TOTAL)	1857.29	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	41	74	41	74	41	74	41	74
Circ. Pl (SUS)	15	15	21	21	29	29	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	-63	-63	77	77	-46	-46	60	60
Long. Q (SUS)	-73	73	289	-289	-59	59	275	-275
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	8	8	-8	-8	-11	-11	11	11
Shear Q (SUS)	22	22	22	22	22	22	22	22
Pm (SUS)	41.0	74.0	41.0	74.0	41.0	74.0	41.0	74.0
Pm+Pl (SUS)	100.3	133.0	98.7	104.1	98.5	130.9	83.4	91.5
Pm+Pl+Q (Total)	182.2	101.6	386.6	288.4	156.6	104.7	358.5	283.8

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	74.00	1203.20	Passed
Pm+Pl (SUS)	132.97	1804.80	Passed
Pm+Pl+Q (TOTAL)	386.60	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: V13"

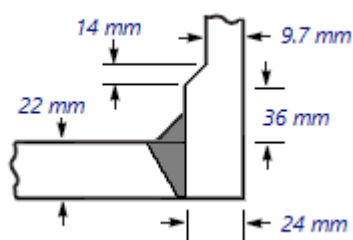
From: 30

Pressure for Reinforcement Calculations	P	28.008	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2025.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		505.00	cm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	36.0000	mm
Height of Beveled Transition	L'	14.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: V13"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.734 in.
Actual Thickness Used in Calculation 0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.01 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.01)$
= 20.4272 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.01 \cdot 36.22 / (1203 \cdot 1.0 - 0.6 \cdot 28.01)$
= 0.8550 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $14.265 / 0.5773$
= 24.7085 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

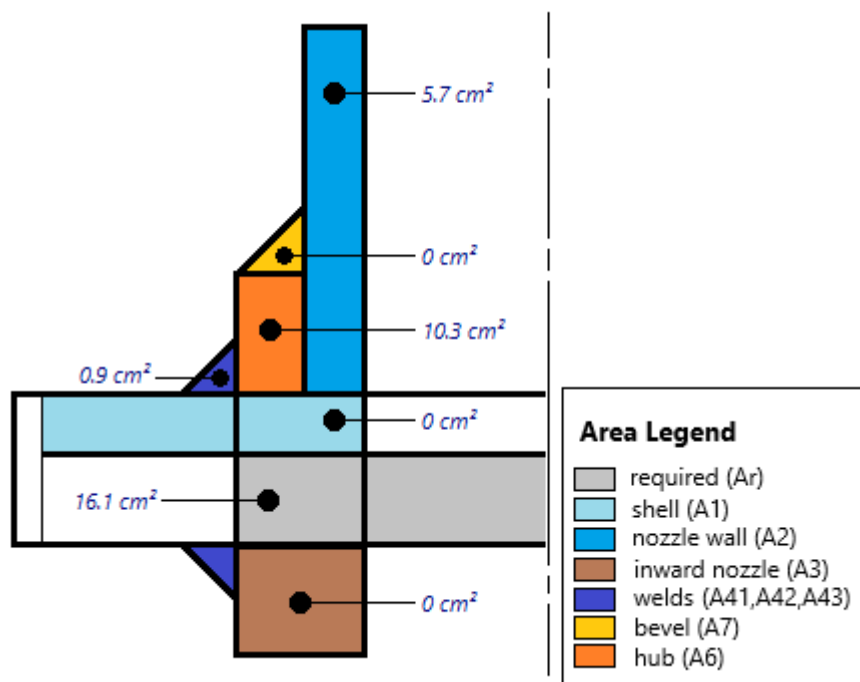
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	144.8618	mm
Parallel to Vessel Wall, opening length	d	72.4309	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	45.2949	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 1203.2 / 1407.1)$
= 0.855

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 1203.2 / 1407.1)$
= 0.855

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.855, 1.0)$
= 0.855



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	16.128	NA	NA
Area in Shell	A1	0.048	NA	NA
Area in Nozzle Wall	A2	5.716	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	10.297	NA	NA
TOTAL AREA AVAILABLE	Atot	16.917	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (72.4309 \cdot 20.4272 \cdot 1.0 + 2 \cdot 22.5 \cdot 20.4272 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 16.128 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 72.431(1.0 \cdot 20.5 - 1.0 \cdot 20.427) - 2 \cdot 22.5$$

$$(1.0 \cdot 20.5 - 1.0 \cdot 20.4272) \cdot (1 - 0.855)$$

$$= 0.048 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 45.29)(8.23 - 0.85)0.8551$$

$$= 5.716 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$$

$$= 0.855 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(45.3, 400.0, 36.0)) \cdot (24.0 - 9.7) \cdot 0.855$$

$$= 10.297 \text{ cm}^2 \text{ Includes Bevel Area of } 1.515 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.3550 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 21.9272 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 21.9272 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 6.3000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.3, \max(21.9272, 3.0)]$$

$$= 6.3000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.355, 6.3)$$

$$= 6.3000 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	343.9,	Allowable	:	1203.2 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2664.1 kgf/cm ²	Passed
Occasional	:	55.3,	Allowable	:	1600.3 kgf/cm ²	Passed
Shear	:	227.6,	Allowable	:	842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 28.01/86.88 = 0.322

Weld Size Calculations, Description: V13"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (16.1278 - 0.048 + 2 \cdot 22.5 \cdot 0.8551 \cdot (1.0 \cdot 20.5 - 20.4272)) \cdot 1407)$
 $= 22665.84 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (16.0137 + 0.8551 - 0.0 \cdot 0.86) \cdot 1407$
 $= 23736.55 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (16.0137 + 0.0 + 0.8551 + (7.888)) \cdot 1407$
 $= 34836.07 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (16.0137 + 0.0 + 0.8551 + 0.0 + (7.888)) \cdot 1407$
 $= 34836.07 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 117.4309 \cdot 10.0 \cdot 0.49 \cdot 1203$
 $= 10875. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 47.4655) \cdot (24.0 - 1.5) \cdot 0.7 \cdot 1203$
 $= 28258. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 117.4309 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 1407$
 $= 39375. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (10875 + 28258) = 39133 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (10875 + 0 + 39375 + 0) = 50250 \text{ kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (10875 + 39375 + 0) = 50250 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 39133	kgf, must exceed W = 22665 kgf	or W1 = 23736 kgf
Path 2-2 = 50250	kgf, must exceed W = 22665 kgf	or W2 = 34836 kgf
Path 3-3 = 50250	kgf, must exceed W = 22665 kgf	or W3 = 34836 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 28 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.7039 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 423.7039 mm

Input Echo, WRC297 Item 1, Description: V13" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2025.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	69.431	mm
Nozzle Thickness	24.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF1	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	28.01	kgf/cm ²
Radial Load	P	-489.47	kgf
Circumferential Shear	Vc	367.10	kgf
Longitudinal Shear	Vl	489.47	kgf
Circumferential Moment	Mc	73.42	kgf-m.
Longitudinal Moment	Ml	95.45	kgf-m.
Torsional Moment	Mt	110.13	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (4.893) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 117.431 + 2 \cdot 1.65 \cdot \sqrt{(1024.25(22.0 - 1.5))}$
 $= 595.614 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2048.500 mm
Nozzle Outside Diameter	(d) =	117.431 mm
Vessel Thickness used	(T) =	20.500 mm
Nozzle Thickness used	(t) =	22.500 mm
T / t	=	0.911
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.573$$

Nr/P	=	0.161
Mr/P	=	0.268



M0/P = 0.091
 N0/P = 0.164
 MrD/Mc = 0.293
 NrDL/Mc = 0.133
 M0d/Mc = 0.113
 N0DL/Mc = -0.081
 MrD/M1 = 0.252
 NrDL/M1 = 0.128
 M0D/M1 = 0.096
 N0DL/M1 = 0.020

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	18	18	18	18
Outplane Bending (P)	63	-63	63	-63
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-111	111	111	-111
Normal Pressure Stress	1371	1399	1371	1399
Outplane Stress Summary	1338	1462	1568	1247

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	18	18	18	18
Inplane Bending (P)	187	-187	187	-187
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-24	-24	24	24
Inplane Bending (ML)	-292	292	292	-292
Inplane Pressure Stress	685	685	685	685
Inplane Stress Summary	574	784	1208	249

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	9	9	-9	-9
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	24	24	24	24
Shear Stress Summary	34	34	15	15

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1340	1464	1569	1247

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	18	18	18	18
Outplane Bending (P)	63	-63	63	-63
Outplane Membrane (Mc)	11	11	-11	-11
Outplane Bending (Mc)	-101	101	101	-101
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



Normal Pressure Stress	685	685	685	685
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Outplane Stress Summary	678	754	857	528
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	18	18	18	18
Inplane Bending (P)	187	-187	187	-187
Inplane Membrane (Mc)	-19	-19	19	19
Inplane Bending (Mc)	-261	261	261	-261
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1371	1399	1371	1399

Inplane Stress Summary	1296	1472	1859	988
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	24	24	24	24

Shear Stress Summary	11	11	37	37
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	1296	1472	1860	991
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	18	18	18	18
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	41	74	41	74

Hoop Stress Summary	56	90	64	97
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	7	7	7	7
Axial Bending (P)	108	-108	108	-108
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-70	-70	70	70
Axial Bending (ML)	-181	181	181	-181
Axial Pressure Stress	20	20	20	20

Axial Stress Summary	-114	29	388	-192
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	8	8	-8	-8
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	22	22	22	22

Shear Stress Summary	31	31	13	13
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	182	103	388	291
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	18	18	18	18
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	11	11	-11	-11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	41	74	41	74

Hoop Stress Summary	72	105	48	81
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	7	7	7	7
Axial Bending (P)	108	-108	108	-108
Axial Membrane (Mc)	-53	-53	53	53
Axial Bending (Mc)	-167	167	167	-167
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	20	20	20	20

Axial Stress Summary	-85	32	358	-194
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-11	-11	11	11
Shear due to Torsion	22	22	22	22

Shear Stress Summary	10	10	34	34
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	158	107	362	285
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1371	1399	1371	1399	1371	1399	1371	1399
Circ. Pl (SUS)	15	15	21	21	-1	-1	37	37
Circ. Q (SUS)	-48	48	174	-174	-74	74	448	-448

Long. Pm (SUS)	685	685	685	685	685	685	685	685
Long. Pl (SUS)	-6	-6	42	42	29	29	7	7
Long. Q (SUS)	-105	105	479	-479	-38	38	164	-164
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	9	9	-9	-9	-12	-12	12	12
Shear Q (SUS)	24	24	24	24	24	24	24	24
Pm (SUS)	1371	1399	1371	1399	1371	1399	1371	1399
Pm+Pl (SUS)	1386	1414	1392	1420	1370	1398	1408	1436
Pm+Pl+Q (Total)	1339	1463	1566	1246	1296	1472	1857	990

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1399.00	1407.13	Passed
Pm+Pl (SUS)	1436.19	2110.70	Passed
Pm+Pl+Q (TOTAL)	1857.29	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	41	74	41	74	41	74	41	74
Circ. Pl (SUS)	15	15	21	21	29	29	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	-63	-63	77	77	-46	-46	60	60
Long. Q (SUS)	-73	73	289	-289	-59	59	275	-275
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	8	8	-8	-8	-11	-11	11	11
Shear Q (SUS)	22	22	22	22	22	22	22	22
Pm (SUS)	41.0	74.0	41.0	74.0	41.0	74.0	41.0	74.0
Pm+Pl (SUS)	100.3	133.0	98.7	104.1	98.5	130.9	83.4	91.5
Pm+Pl+Q (Total)	182.2	101.6	386.6	288.4	156.6	104.7	358.5	283.8

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	74.00	1203.20	Passed
Pm+Pl (SUS)	132.97	1804.80	Passed
Pm+Pl+Q (TOTAL)	386.60	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: T12"

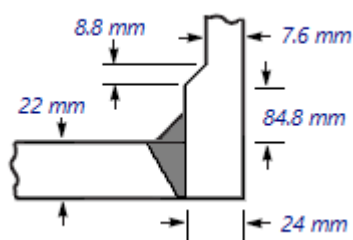
From: 30

Pressure for Reinforcement Calculations	P	28.107	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2025.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		405.00	cm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	84.8118	mm
Height of Beveled Transition	L'	8.8118	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: T12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.11 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.11)$
 $= 20.4999$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.11 \cdot 24.02 / (1203 \cdot 1.0 - 0.6 \cdot 28.11)$
 $= 0.5690$ mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot$ Corroded Hub Thickness
 $= 2.5 \cdot 22.5$, less than the hub height, use (e-2)
 $= 56.2500$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	134.0342	mm
Parallel to Vessel Wall	Rn+tn+t	67.0171	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: T12".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0690	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 21.9999	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 21.9999	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(22.0, 3.0)]$
 $= 4.9200$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.069, 4.92)$
 $= 4.9200$ mm

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 428.5,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2579.5 kgf/cm ²	Passed
Occasional	: 48.7,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 269.2,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :



Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: T12"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0691 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 423.0691 mm

Input Echo, WRC297 Item 1, Description: T12" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2025.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1407.13 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13 kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	45.034	mm
Nozzle Thickness	24.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF1	
Nozzle Cold S.I. Allowable	SNmc	1203.20 kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1203.20 kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	28.11	kgf/cm ²
Radial Load	P	-285.53	kgf
Circumferential Shear	Vc	214.15	kgf
Longitudinal Shear	Vl	285.53	kgf
Circumferential Moment	Mc	37.10	kgf-m.
Longitudinal Moment	Ml	37.10	kgf-m.
Torsional Moment	Mt	42.81	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (4.229) is < 5
 check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (3.876) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:



Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 93.034 + 2 \cdot 1.65 \cdot \sqrt{(1024.25(22.0 - 1.5))}$$

$$= 571.217 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2048.500 mm
Nozzle Outside Diameter	(d) =	93.034 mm
Vessel Thickness used	(T) =	20.500 mm
Nozzle Thickness used	(t) =	22.500 mm
T / t	=	0.911
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.454$$

Nr/P	=	0.176
Mr/P	=	0.302
M0/P	=	0.103
N0/P	=	0.180
MrD/Mc	=	0.295
NrDL/Mc	=	0.131
M0d/Mc	=	0.115
N0DL/Mc	=	-0.093
MrD/Ml	=	0.268
NrDL/Ml	=	0.130
M0D/Ml	=	0.102
N0DL/Ml	=	-0.010

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	41	-41	41	-41
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-58	58	58	-58
Normal Pressure Stress	1376	1404	1376	1404
Outplane Stress Summary	1373	1433	1487	1315

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	11	11	11	11
Inplane Bending (P)	122	-122	122	-122
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-12	-12	12	12
Inplane Bending (ML)	-152	152	152	-152
Inplane Pressure Stress	688	688	688	688
Inplane Stress Summary	658	717	988	436

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	7	7	-7	-7
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	15	15	15	15
Shear Stress Summary	22	22	8	8

**Vessel Stresses (kgf/cm²):**

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1373	1434	1487	1315

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	41	-41	41	-41
Outplane Membrane (Mc)	8	8	-8	-8
Outplane Bending (Mc)	-65	65	65	-65
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	688	688	688	688
Outplane Stress Summary	685	732	798	584

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	11	11	11	11
Inplane Bending (P)	122	-122	122	-122
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-167	167	167	-167
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1376	1404	1376	1404
Inplane Stress Summary	1330	1448	1691	1137

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-9	-9	9	9
Torsional Shear (Mt)	15	15	15	15
Shear Stress Summary	5	5	24	24

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1330	1448	1692	1139

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	12	12	12	12
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	24	58	24	58
Hoop Stress Summary	37	71	35	69



Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	5	5	5	5
Axial Bending (P)	72	-72	72	-72
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-50	-50	50	50
Axial Bending (ML)	-96	96	96	-96
Axial Pressure Stress	12	12	12	12
Axial Stress Summary	-56	-8	236	-100

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	6	6	-6	-6
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	13	13	13	13
Shear Stress Summary	20	20	7	7

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	102	89	237	170

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	12	12	12	12
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	8	8	-8	-8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	24	58	24	58
Hoop Stress Summary	45	79	27	61

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	5	5	5	5
Axial Bending (P)	72	-72	72	-72
Axial Membrane (Mc)	-50	-50	50	50
Axial Bending (Mc)	-108	108	108	-108
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	12	12	12	12
Axial Stress Summary	-68	3	249	-112

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-8	-8	8	8
Shear due to Torsion	13	13	13	13

Shear Stress Summary 5 5 22 22

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	114	79	251	179

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1376	1404	1376	1404	1376	1404	1376	1404
Circ. Pl (SUS)	12	12	12	12	-1	-1	23	23
Circ. Q (SUS)	-17	17	99	-99	-45	45	289	-289
Long. Pm (SUS)	688	688	688	688	688	688	688	688
Long. Pl (SUS)	-1	-1	23	23	20	20	4	4
Long. Q (SUS)	-30	30	274	-274	-24	24	106	-106
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-9	-9	9	9
Shear Q (SUS)	15	15	15	15	15	15	15	15
Pm (SUS)	1376	1404	1376	1404	1376	1404	1376	1404
Pm+Pl (SUS)	1388	1416	1388	1416	1375	1403	1399	1427
Pm+Pl+Q (Total)	1371	1433	1487	1317	1330	1448	1688	1139

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1404.00	1407.13	Passed
Pm+Pl (SUS)	1427.11	2110.70	Passed
Pm+Pl+Q (TOTAL)	1688.65	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	24	58	24	58	24	58	24	58
Circ. Pl (SUS)	12	12	12	12	20	20	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. Pl (SUS)	-45	-45	55	55	-45	-45	55	55
Long. Q (SUS)	-24	24	168	-168	-36	36	180	-180
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	6	6	-6	-6	-8	-8	8	8
Shear Q (SUS)	13	13	13	13	13	13	13	13
Pm (SUS)	24.0	58.0	24.0	58.0	24.0	58.0	24.0	58.0
Pm+Pl (SUS)	70.0	103.7	68.1	74.7	78.6	112.1	68.6	72.9
Pm+Pl+Q (Total)	100.5	87.7	235.2	171.6	113.4	78.3	249.0	180.0



Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	58.00	1203.20	Passed
Pm+Pl (SUS)	112.15	1804.80	Passed
Pm+Pl+Q (TOTAL)	249.00	3609.61	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: L12"

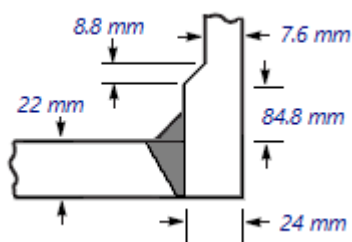
From: 30

Pressure for Reinforcement Calculations	P	28.020	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2025.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	500.0000	mm
Distance from Bottom/Left Tangent		455.00	cm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		28.90	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	84.8118	mm
Height of Beveled Transition	L'	8.8118	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	179.4260	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: L12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.773 in.
Actual Thickness Used in Calculation	0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.02 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.02)$
 $= 20.4357 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.02 \cdot 24.02 / (1203 \cdot 1.0 - 0.6 \cdot 28.02)$
 $= 0.5672 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 22.5$, less than the hub height, use (e-2)
 $= 56.2500 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	141.0510 mm
Parallel to Vessel Wall	Rn+tn+t	70.5255 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L12".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Weld Size Calculations, Description: L12"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: L12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.773 in.
Actual Thickness Used in Calculation	0.301 in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.02 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.02)$
= 20.4357 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.02 \cdot 24.02 / (1203 \cdot 1.0 - 0.6 \cdot 28.02)$
= 0.5672 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
= 2.5 • Corroded Hub Thickness
= 2.5 • 22.5, less than the hub height, use (e-2)
= 56.2500 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	134.0342	mm
Parallel to Vessel Wall	Rn+tn+t	67.0171	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L12".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0672	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 21.9357	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 21.9357	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[4.92, max(21.9357, 3.0)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.0672, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Weld Size Calculations, Description: L12"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 28.0943 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 232.7344 mm

*This nozzle or its diameter limit intersects weld seam # [3] on this element. The
distance between the nozzle and seam is, 18.77 mm , while the distance from the center
of the nozzle to the edge of the diameter limit is: 67.02 mm*

Input, Nozzle Desc: L22"

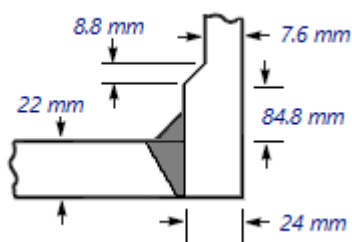
From: 30

Pressure for Reinforcement Calculations	P	28.077	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2025.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	500.0000	mm
Distance from Bottom/Left Tangent		455.00	cm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-28.90	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	84.8118	mm
Height of Beveled Transition	L'	8.8118	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	179.4260	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: L22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.773 in.
Actual Thickness Used in Calculation	0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.08 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.08)$
 $= 20.4775 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.08 \cdot 24.02 / (1203 \cdot 1.0 - 0.6 \cdot 28.08)$
 $= 0.5684 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 22.5$, less than the hub height, use (e-2)
 $= 56.2500 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	141.0506 mm
Parallel to Vessel Wall	Rn+tn+t	70.5253 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L22".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Weld Size Calculations, Description: L22"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: L22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.773 in.
Actual Thickness Used in Calculation	0.301 in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.08 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.08)$
= 20.4775 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.08 \cdot 24.02 / (1203 \cdot 1.0 - 0.6 \cdot 28.08)$
= 0.5684 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
= 2.5 • Corroded Hub Thickness
= 2.5 • 22.5, less than the hub height, use (e-2)
= 56.2500 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	134.0342	mm
Parallel to Vessel Wall	Rn+tn+t	67.0171	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L22".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0684	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 21.9775	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 21.9775	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[4.92, max(21.9775, 3.0)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.0684, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Weld Size Calculations, Description: L22"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 28.0943 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 232.7344 mm

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Input, Nozzle Desc: B26"

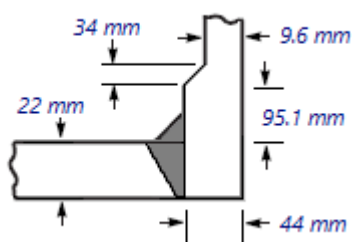
From: 40

Pressure for Reinforcement Calculations	P	28.107	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2025.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		755.00	cm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	95.0683	mm
Height of Beveled Transition	L'	34.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		44.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: B26"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 5.869 in.
Actual Thickness Used in Calculation 0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.11 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.11)$
= 20.4999 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.11 \cdot 76.04 / (1203 \cdot 1.0 - 0.6 \cdot 28.11)$
= 1.8015 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $34.399 / 0.5773$
= 59.5805 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

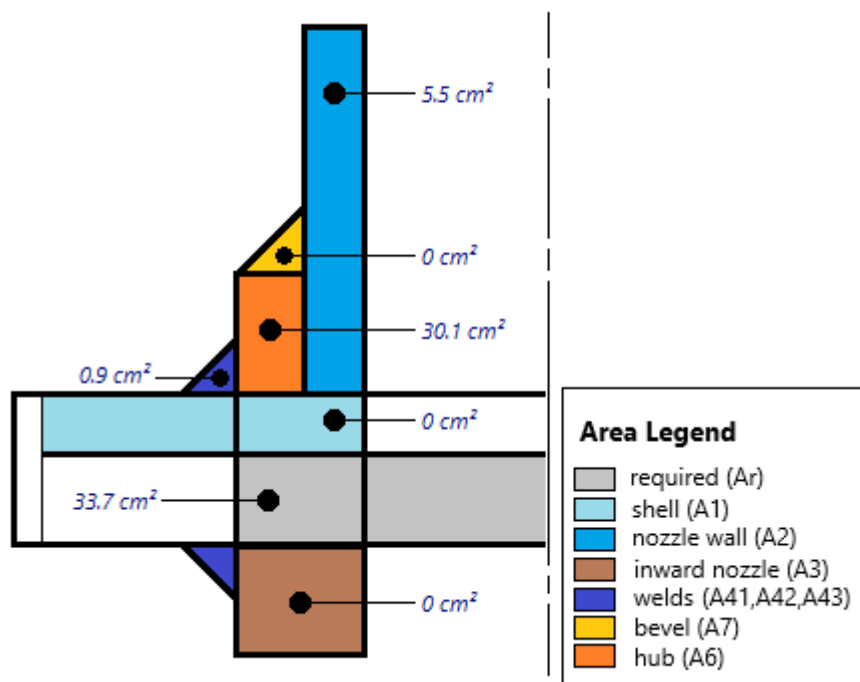
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	304.1452	mm
Parallel to Vessel Wall, opening length	d	152.0726	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 1203.2 / 1407.1)$
= 0.855

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 1203.2 / 1407.1)$
= 0.855

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.855, 1.0)$
= 0.855



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	33.700	NA	NA
Area in Shell	A1	0.000	NA	NA
Area in Nozzle Wall	A2	5.521	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	30.149	NA	NA
TOTAL AREA AVAILABLE	Atot	36.525	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (152.0726 \cdot 20.4999 \cdot 1.0 + 2 \cdot 42.5 \cdot 20.4999 \cdot 1.0 \cdot (1 - 0.86))$
 $= 33.700 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 152.073(1.0 \cdot 20.5 - 1.0 \cdot 20.5) - 2 \cdot 42.5$
 $(1.0 \cdot 20.5 - 1.0 \cdot 20.4999) \cdot (1 - 0.855)$
 $= 0.000 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 51.25)(8.1 - 1.8)0.8551$
 $= 5.521 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$
 $= 0.855 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(51.2, 200.0, 95.1)) \cdot (44.0 - 9.6) \cdot 0.855$$

$$= 30.149 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.3015 \text{ mm}$
 Wall Thickness per UG16(b), $tr_{16b} = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb_1 = 21.9999 \text{ mm}$
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 21.9999 \text{ mm}$
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb_3 = 7.7200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[7.72, \max(22.0, 3.0)]$$

$$= 7.7200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.3015, 7.72)$$

$$= 7.7200 \text{ mm}$$

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	410.4,	Allowable	:	1203.2 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2597.6 kgf/cm ²	Passed
Occasional	:	125.2,	Allowable	:	1600.3 kgf/cm ²	Passed
Shear	:	212.9,	Allowable	:	842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-57 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $28.11/52.01 = 0.540$

Weld Size Calculations, Description: B26"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
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Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------

Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (33.7001 - 0.0001 + 2 \cdot 42.5 \cdot 0.8551 \cdot (1.0 \cdot 20.5 - 20.4999)) \cdot 1407)$
 $= 47420.48 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (35.6702 + 0.8551 - 0.0 \cdot 0.86) \cdot 1407$
 $= 51395.89 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (35.6702 + 0.0 + 0.8551 + (14.8996)) \cdot 1407$
 $= 72361.65 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (35.6702 + 0.0 + 0.8551 + 0.0 + (14.8996)) \cdot 1407$
 $= 72361.65 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 237.0726 \cdot 10.0 \cdot 0.49 \cdot 1203$
 $= 21955. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 97.2863) \cdot (44.0 - 1.5) \cdot 0.7 \cdot 1203$
 $= 109402. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 237.0726 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 1407$
 $= 79492. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (21955 + 109402) = 131357 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (21955 + 0 + 79492 + 0) = 101447 \text{ kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (21955 + 79492 + 0) = 101447 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 131357 kgf, must exceed W = 47420 kgf	or W1 = 51395 kgf
Path 2-2 = 101446 kgf, must exceed W = 47420 kgf	or W2 = 72361 kgf
Path 3-3 = 101446 kgf, must exceed W = 47420 kgf	or W3 = 72361 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 28 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.



The Drop for this Nozzle is : 6.9626 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.9626 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 133.06 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 152.07 mm

Input Echo, WRC297 Item 1, Description: B26" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2025.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	149.073	mm
Nozzle Thickness	44.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF1	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	28.11	kgf/cm ²
Radial Load	P	-978.93	kgf
Circumferential Shear	Vc	734.20	kgf
Longitudinal Shear	Vl	978.93	kgf
Circumferential Moment	Mc	293.68	kgf-m.
Longitudinal Moment	Ml	381.78	kgf-m.
Torsional Moment	Mt	440.52	kgf-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

*Warning - The ratio, Dn/Tn (5.388) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 237.073 + 2 \cdot 1.65 \cdot \sqrt{(1024.25(22.0 - 1.5))}$$

$$= 715.255 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2048.500 mm
Nozzle Outside Diameter	(d) =	237.073 mm
Vessel Thickness used	(T) =	20.500 mm
Nozzle Thickness used	(t) =	42.500 mm
T / t	=	0.482
d / t	=	10.000



$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.157$$

Nr/P	=	0.097
Mr/P	=	0.204
M0/P	=	0.060
N0/P	=	0.125
MrD/Mc	=	0.305
NrDL/Mc	=	0.092
M0d/Mc	=	0.095
N0DL/Mc	=	0.046
MrD/M1	=	0.177
NrDL/M1	=	0.085
M0D/M1	=	0.057
N0DL/M1	=	0.126

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	29	29	29	29
Outplane Bending (P)	83	-83	83	-83
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-48	-48	48	48
Outplane Bending (ML)	-131	131	131	-131
Normal Pressure Stress	1376	1404	1376	1404
Outplane Stress Summary	1309	1432	1668	1267

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	22	22	22	22
Inplane Bending (P)	285	-285	285	-285
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-32	-32	32	32
Inplane Bending (ML)	-407	407	407	-407
Inplane Pressure Stress	688	688	688	688
Inplane Stress Summary	556	799	1435	50

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	9	9	-9	-9
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	24	24	24	24
Shear Stress Summary	33	33	14	14

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1310	1434	1669	1267

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	29	29	29	29
Outplane Bending (P)	83	-83	83	-83



Outplane Membrane (Mc)	-13	-13	13	13
Outplane Bending (Mc)	-167	167	167	-167
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	688	688	688	688

Outplane Stress Summary	619	787	982	479
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	22	22	22	22
Inplane Bending (P)	285	-285	285	-285
Inplane Membrane (Mc)	-27	-27	27	27
Inplane Bending (Mc)	-539	539	539	-539
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1376	1404	1376	1404

Inplane Stress Summary	1117	1653	2250	629
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	24	24	24	24

Shear Stress Summary	11	11	37	37
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	1118	1653	2251	638
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	29	29	29	29
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-48	-48	48	48
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	47	80	47	80

Hoop Stress Summary	27	61	124	158
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	3	3	3	3
Axial Bending (P)	50	-50	50	-50
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-35	-35	35	35
Axial Bending (ML)	-71	71	71	-71
Axial Pressure Stress	23	23	23	23

Axial Stress Summary	-29	13	184	-60
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**Nozzle Stresses (kgf/cm²):**

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11
Shear Stress Summary	16	16	7	7

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	65	66	185	218

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	29	29	29	29
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-13	-13	13	13
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	47	80	47	80
Hoop Stress Summary	62	96	89	123

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	50	-50	50	-50
Axial Membrane (Mc)	-26	-26	26	26
Axial Bending (Mc)	-106	106	106	-106
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	23	23	23	23
Axial Stress Summary	-55	56	211	-102

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-6	-6	6	6
Shear due to Torsion	11	11	11	11
Shear Stress Summary	5	5	17	17

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	118	96	213	229

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	1376	1404	1376	1404	1376	1404	1376	1404
Circ. Pl (SUS)	-19	-19	77	77	-5	-5	49	49
Circ. Q (SUS)	-48	48	214	-214	-254	254	824	-824
Long. Pm (SUS)	688	688	688	688	688	688	688	688
Long. Pl (SUS)	-10	-10	54	54	16	16	42	42
Long. Q (SUS)	-122	122	692	-692	-84	84	250	-250
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	9	9	-9	-9	-12	-12	12	12
Shear Q (SUS)	24	24	24	24	24	24	24	24
Pm (SUS)	1376	1404	1376	1404	1376	1404	1376	1404
Pm+Pl (SUS)	1357	1385	1453	1481	1371	1399	1425	1453
Pm+Pl+Q (Total)	1310	1434	1667	1267	1117	1653	2250	637

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1404.00	1407.13	Passed
Pm+Pl (SUS)	1481.11	2110.70	Passed
Pm+Pl+Q (TOTAL)	2250.02	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	47	80	47	80	47	80	47	80
Circ. Pl (SUS)	-19	-19	77	77	16	16	42	42
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. Pl (SUS)	-32	-32	38	38	-23	-23	29	29
Long. Q (SUS)	-21	21	121	-121	-56	56	156	-156
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-6	-6	6	6
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	47.0	80.0	47.0	80.0	47.0	80.0	47.0	80.0
Pm+Pl (SUS)	37.9	70.5	124.3	157.2	64.1	96.7	89.9	122.5
Pm+Pl+Q (Total)	65.3	65.2	182.8	217.5	119.4	96.6	210.4	228.5

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	80.00	1203.20	Passed
Pm+Pl (SUS)	157.17	1804.80	Passed
Pm+Pl+Q (TOTAL)	228.54	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: B13"

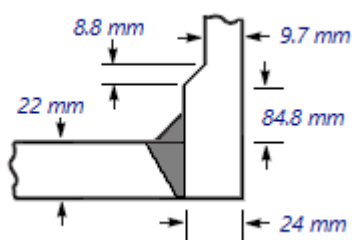
From: 40

Pressure for Reinforcement Calculations	P	28.008	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2025.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		755.00	cm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	
Hub Height of Integral Nozzle	h	84.8118	mm
Height of Beveled Transition	L'	8.8118	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: B13"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.734 in.
 Actual Thickness Used in Calculation 0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.01 \cdot 1014.0 / (1407 \cdot 1.0 - 0.6 \cdot 28.01)$
 $= 20.4272 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.01 \cdot 36.22 / (1203 \cdot 1.0 - 0.6 \cdot 28.01)$
 $= 0.8550 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 22.5$, less than the hub height, use (e-2)
 $= 56.2500 \text{ mm}$

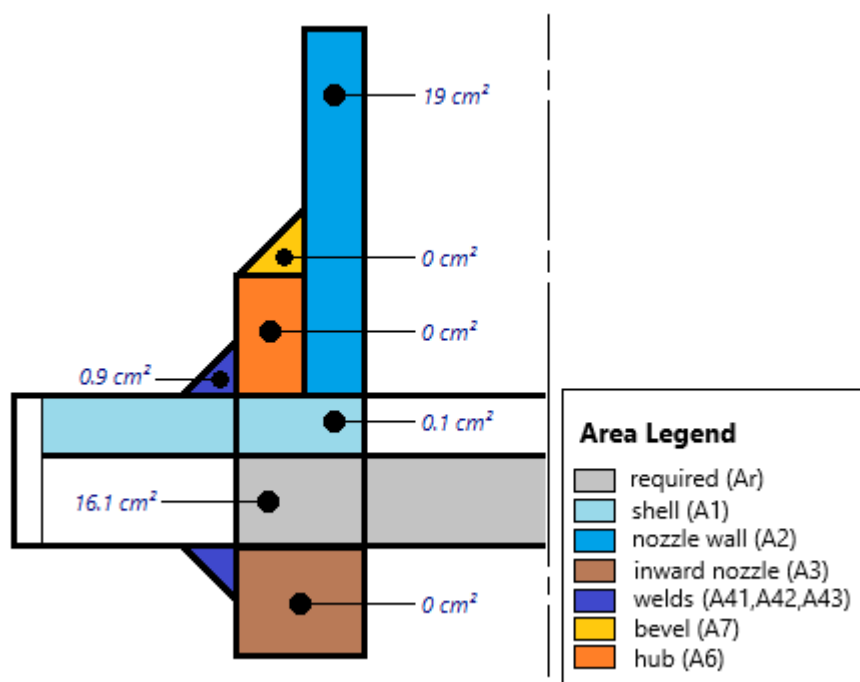
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4309	mm
Parallel to Vessel Wall	Rn+tn+t	79.2154	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n/S_v)$
 $= \min(1, 1203.2/1407.1)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n/S_v)$
 $= \min(1, 1203.2/1407.1)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	16.128	NA	NA
Area in Shell	A1	0.058	NA	NA
Area in Nozzle Wall	A2	18.971	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	19.884	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (72.4309 \cdot 20.4272 \cdot 1.0 + 2 \cdot 22.5 \cdot 20.4272 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 16.128 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 86.0(1.0 \cdot 20.5 - 1.0 \cdot 20.427) - 2 \cdot 22.5$$

$$(1.0 \cdot 20.5 - 1.0 \cdot 20.4272) \cdot (1 - 0.855)$$

$$= 0.058 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 51.25)(22.5 - 0.85)0.8551$$

$$= 18.971 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$$

$$= 0.855 \text{ cm}^2$$

There are no hub area calculations because Figure UG-40 (e-2) was used.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.3550 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 21.9272 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 21.9272 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 6.3000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.3, \max(21.9272, 3.0)]$$

$$= 6.3000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.355, 6.3)$$

$$= 6.3000 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	343.9,	Allowable	:	1203.2 kgf/cm²	Passed
Expansion	:	0.0,	Allowable	:	2664.1 kgf/cm²	Passed



Occasional : 55.3, Allowable : 1600.3 kgf/cm² Passed
 Shear : 227.6, Allowable : 842.2 kgf/cm² Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 9.735, tr = 0.855, c = 1.5 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.104, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: B13"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (16.1278 - 0.0579 + 2 \cdot 22.5 \cdot 0.8551 \cdot (1.0 \cdot 20.5 - 20.4272)) \cdot 1407)$
 $= 22651.94 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (18.9708 + 0.8551 - 0.0 \cdot 0.86) \cdot 1407$
 $= 27897.58 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot (\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v$
 $= (18.9708 + 0.0 + 0.8551 + (7.888)) \cdot 1407$
 $= 38997.10 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+(2 \cdot (\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v$
 $= (18.9708 + 0.0 + 0.8551 + 0.0) \cdot 1407$
 $= 38997.10 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 117.4309 \cdot 10.0 \cdot 0.49 \cdot 1203$
 $= 10875. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$

$$= (3.1416 \cdot 47.4655) \cdot (24.0 - 1.5) \cdot 0.7 \cdot 1203$$

$$= 28258. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 117.4309 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 1407$$

$$= 39375. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (10875 + 28258) = 39133 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (10875 + 0 + 39375 + 0) = 50250 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (10875 + 39375 + 0) = 50250 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 39133	kgf, must exceed W = 22651 kgf	or W1 = 27897 kgf
Path 2-2 = 50250	kgf, must exceed W = 22651 kgf	or W2 = 38997 kgf
Path 3-3 = 50250	kgf, must exceed W = 22651 kgf	or W3 = 38997 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 28 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.7039 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 423.7039 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 79.22 mm

Input Echo, WRC297 Item 1, Description: B13" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2025.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1407.13 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13 kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	69.431	mm
Nozzle Thickness	24.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF1	
Nozzle Cold S.I. Allowable	SNmc	1203.20 kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1203.20 kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.



Design Internal Pressure	Dp	28.01	kgf/cm ²
Radial Load	P	-489.47	kgf
Circumferential Shear	Vc	367.10	kgf
Longitudinal Shear	Vl	489.47	kgf
Circumferential Moment	Mc	73.42	kgf-m.
Longitudinal Moment	Ml	95.45	kgf-m.
Torsional Moment	Mt	110.13	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (4.893) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 117.431 + 2 \cdot 1.65 \cdot \sqrt{(1024.25(22.0 - 1.5))}$$

$$= 595.614 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2048.500 mm
Nozzle Outside Diameter	(d) =	117.431 mm
Vessel Thickness used	(T) =	20.500 mm
Nozzle Thickness used	(t) =	22.500 mm
T / t	=	0.911
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.573$$

Nr/P	=	0.161
Mr/P	=	0.268
M0/P	=	0.091
N0/P	=	0.164
MrD/Mc	=	0.293
NrDL/Mc	=	0.133
M0d/Mc	=	0.113
N0DL/Mc	=	-0.081
MrD/Ml	=	0.252
NrDL/Ml	=	0.128
M0D/Ml	=	0.096
N0DL/Ml	=	0.020

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	18	18	18	18
Outplane Bending (P)	63	-63	63	-63
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-3	-3	3	3
Outplane Bending (Ml)	-111	111	111	-111
Normal Pressure Stress	1371	1399	1371	1399
Outplane Stress Summary	1338	1462	1568	1247

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	18	18	18	18
Inplane Bending (P)	187	-187	187	-187
Inplane Membrane (Mc)	0	0	0	0



Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-24	-24	24	24
Inplane Bending (ML)	-292	292	292	-292
Inplane Pressure Stress	685	685	685	685

Inplane Stress Summary	574	784	1208	249
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	9	9	-9	-9
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	24	24	24	24

Shear Stress Summary	34	34	15	15
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	1340	1464	1569	1247
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	18	18	18	18
Outplane Bending (P)	63	-63	63	-63
Outplane Membrane (Mc)	11	11	-11	-11
Outplane Bending (Mc)	-101	101	101	-101
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	685	685	685	685

Outplane Stress Summary	678	754	857	528
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	18	18	18	18
Inplane Bending (P)	187	-187	187	-187
Inplane Membrane (Mc)	-19	-19	19	19
Inplane Bending (Mc)	-261	261	261	-261
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1371	1399	1371	1399

Inplane Stress Summary	1296	1472	1859	988
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	24	24	24	24

Shear Stress Summary	11	11	37	37
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



Two * Max Shear Stress	1296	1472	1860	991
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
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Hoop Membrane (P)	18	18	18	18
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	41	74	41	74

Hoop Stress Summary	56	90	64	97
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
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Axial Membrane (P)	7	7	7	7
Axial Bending (P)	108	-108	108	-108
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-70	-70	70	70
Axial Bending (ML)	-181	181	181	-181
Axial Pressure Stress	20	20	20	20

Axial Stress Summary	-114	29	388	-192
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	B1 Inside
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Shear due to (Vc)	8	8	-8	-8
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	22	22	22	22

Shear Stress Summary	31	31	13	13
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	B1 Inside
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Two * Max Shear Stress	182	103	388	291
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
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Hoop Membrane (P)	18	18	18	18
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	11	11	-11	-11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	41	74	41	74

Hoop Stress Summary	72	105	48	81
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
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Axial Membrane (P)	7	7	7	7
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Axial Bending (P)	108	-108	108	-108
Axial Membrane (Mc)	-53	-53	53	53
Axial Bending (Mc)	-167	167	167	-167
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	20	20	20	20

Axial Stress Summary	-85	32	358	-194
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-11	-11	11	11
Shear due to Torsion	22	22	22	22
Shear Stress Summary	10	10	34	34

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	158	107	362	285

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1371	1399	1371	1399	1371	1399	1371	1399
Circ. Pl (SUS)	15	15	21	21	-1	-1	37	37
Circ. Q (SUS)	-48	48	174	-174	-74	74	448	-448
Long. Pm (SUS)	685	685	685	685	685	685	685	685
Long. Pl (SUS)	-6	-6	42	42	29	29	7	7
Long. Q (SUS)	-105	105	479	-479	-38	38	164	-164
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	9	9	-9	-9	-12	-12	12	12
Shear Q (SUS)	24	24	24	24	24	24	24	24
Pm (SUS)	1371	1399	1371	1399	1371	1399	1371	1399
Pm+Pl (SUS)	1386	1414	1392	1420	1370	1398	1408	1436
Pm+Pl+Q (Total)	1339	1463	1566	1246	1296	1472	1857	990

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1399.00	1407.13	Passed
Pm+Pl (SUS)	1436.19	2110.70	Passed
Pm+Pl+Q (TOTAL)	1857.29	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	41	74	41	74	41	74	41	74
Circ. Pl (SUS)	15	15	21	21	29	29	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	-63	-63	77	77	-46	-46	60	60
Long. Q (SUS)	-73	73	289	-289	-59	59	275	-275
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	8	8	-8	-8	-11	-11	11	11
Shear Q (SUS)	22	22	22	22	22	22	22	22
Pm (SUS)	41.0	74.0	41.0	74.0	41.0	74.0	41.0	74.0
Pm+Pl (SUS)	100.3	133.0	98.7	104.1	98.5	130.9	83.4	91.5
Pm+Pl+Q (Total)	182.2	101.6	386.6	288.4	156.6	104.7	358.5	283.8

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	74.00	1203.20	Passed
Pm+Pl (SUS)	132.97	1804.80	Passed
Pm+Pl+Q (TOTAL)	386.60	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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